

6. Animal Classification

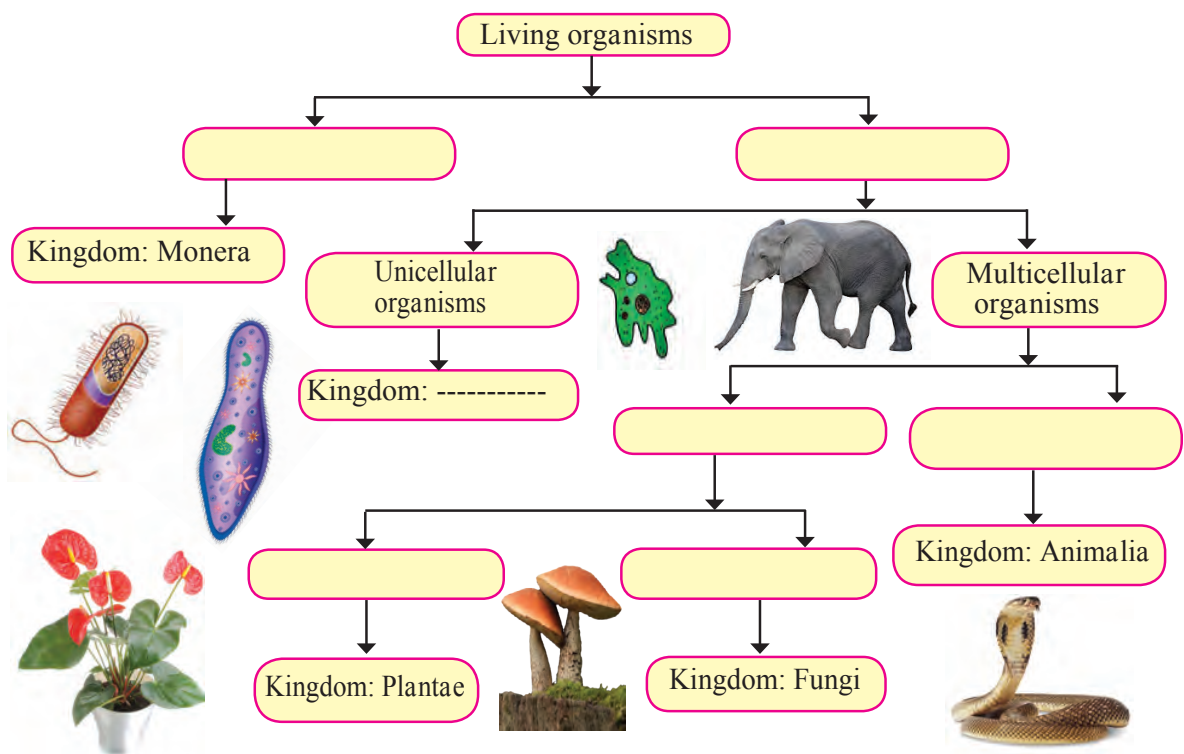


- History of animal classification
- New (method) system of animal classification.
- Kingdom- Animalia.



Can you recall? Which criteria are used for classification of organisms?

You have studied the classification of living organisms in earlier classes. The living organisms present around us are mainly plants and animals. We have studied the criteria of their classification. With the help of that, complete the following chart.



6.1 Classification of living organisms



Can you recall? How the plants are classified?

We have studied the plants classification in last year. It helped us to understand the diversity of plants around us.

You must be looking varieties of animals around you. Some animals are too small whereas some are too big. Some animals are terrestrial where as some are aquatic. Some animals crawl on land, some swim in water whereas some fly in air. Some animals have scales on skin whereas some have feathers or hairs. In this way, there is huge diversity among the animals too. According to recent studies, estimated number of animal species on earth is approximately 7 million. It is impossible to study each and every species. However, if groups and sub-groups of animals are formed depending upon the similarities & differences, it will make it very easy to study such vast variety of animals.

Formation of groups and sub-groups of animals depending upon similarities and differences among animals is called as animal classification.

History of animal classification

Time to time, different scientists have tried to classify the animals. Greek philosopher Aristotle was the first to perform the animal classification. Aristotle classified the animals according to the criteria like body size, habits and habitats. Further, as per the new developments in sciences, references were changed and thereby the criteria of animal classification too. Classification proposed by Aristotle is known as 'Artificial method'. Besides Aristotle, artificial method of classification was followed by Theophrastus, Pliny, John Ray, Linnaeus, etc. Later on, 'Natural system of classification' was followed. Natural system of classification was based on various criteria like body organization, types of cells, chromosomes, bio-chemical properties, etc. By the time, system of classification based on evolution was also brought into practice. It was used by Dobzhansky and Meyer. Recently, Carl Woese has also proposed the animal classification

Benefits of animal classification

1. Study of animals becomes convenient.
2. Study of few animals from a group helps to understand about that entire animal group.
3. It gives idea about animal evolution.
4. Animals can be easily identified with great accuracy.
5. It helps to understand the relationship of animals with other living organisms.
6. It helps to understand the habitat of each animal and its exact role in the nature.
7. It helps to understand various adaptations shown by animals.

Traditional method of animal classification

Traditionally, depending upon presence or absence of the notochord, the animal kingdom has been divided into two groups- **Non-chordates and Chordates**.

A. Non-Chordates : Characters of non-chordate animals are as follows

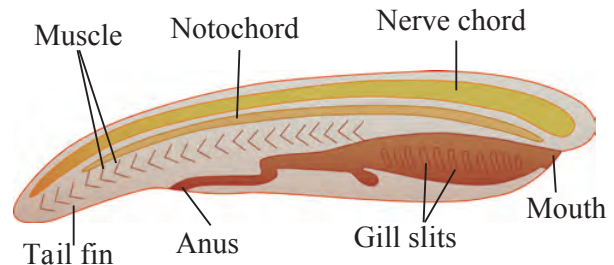
1. Body is not supported by rod-like notochord.
2. Pharyngeal gill-slits are absent.
3. Nerve cord; if present, it is on ventral side. It is solid & paired.
4. Heart, if present, it is on dorsal side.

Non-chordates are classified / divided into ten phyla. Those phyla are- Protozoa, Porifera, Coelenterata / Cnidaria, Platyhelminthes, Aschelminthes, Annelida, Arthropoda, Mollusca, Echinodermata and Hemichordata

Chordates : Characters of chordates are as follows

1. Body is supported by notochord.
2. Pharyngeal gill-slits or lungs are present for respiration.
3. Nerve cord is present on dorsal side of body. It is hollow.
4. Heart is present on ventral side of body.

Notochord is a long rod like supporting structure present on dorsal side of animal body. It keeps the nerve tissue isolated from remaining body.

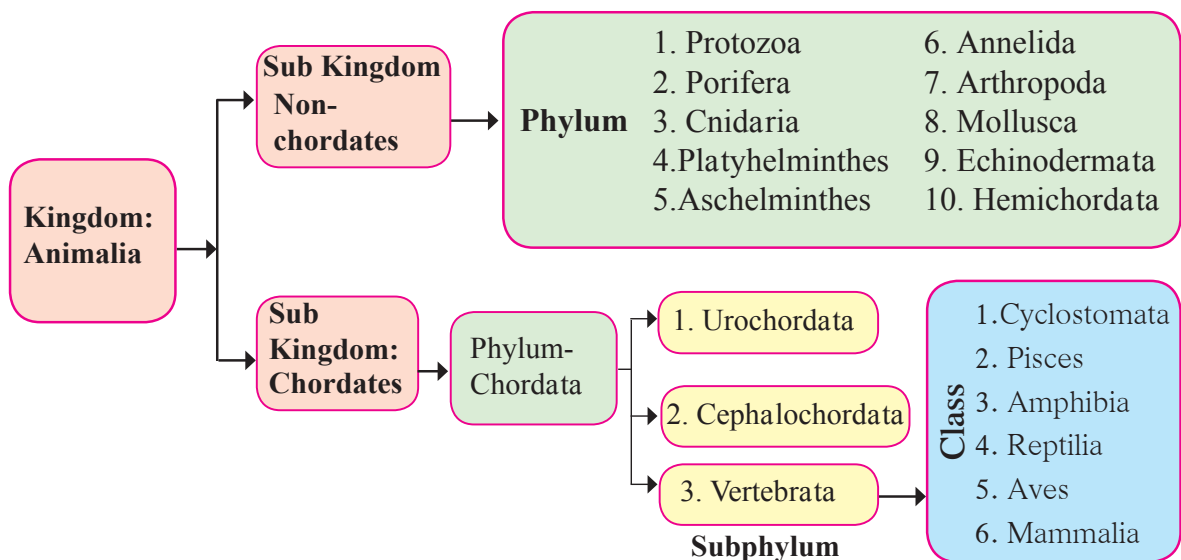


6.2 Characteristics of Chordates



Do you know?

All chordate animals are grouped together in a single phylum and the name of phylum is same i.e. Phylum- Chordata. This phylum has been divided into three subphyla as- Urochordata, Cephalochordata & Vertebrata. Sub-phylum Vertebrata has been further divided into six classes as- Class: Cyclostomata, Class: Pisces, Class: Amphibia, Class: Reptilia, Class: Aves and Class: Mammalia.



6.3 Conventional System of Animal Classification

This system of animal classification was in practice till now. However, now a days, new system of classification is followed. We will study this new system of animal classification in brief.

At present, according to the five kingdom classification system of Robert Whittaker, all multicellular animals are included in Kingdom: Animalia. This system of classification is based upon some criteria like Body organization, Body symmetry, Body cavity, Germinal layers, Segmentation, etc.

Criteria for new system of classification

A. Grades of organization

Body of animals is made up of cells. In case of multicellular animals, many cells are performing different functions in their body.

In case of multicellular animals, if tissues are not formed, their body organization is called as 'Cellular grade organization'. Ex. Phylum-Porifera.

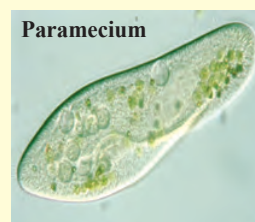
In case of some animals, cells come together to form tissues with the help of which all the body functions are performed. Such animals show 'Cell - tissue grade' organization. Ex. Animals from phylum- Cnidaria.

Flat worms show 'Tissue-Organ grade' organization. In this type of organization, tissues are organized to form some organs. However, complete organ systems are not formed.

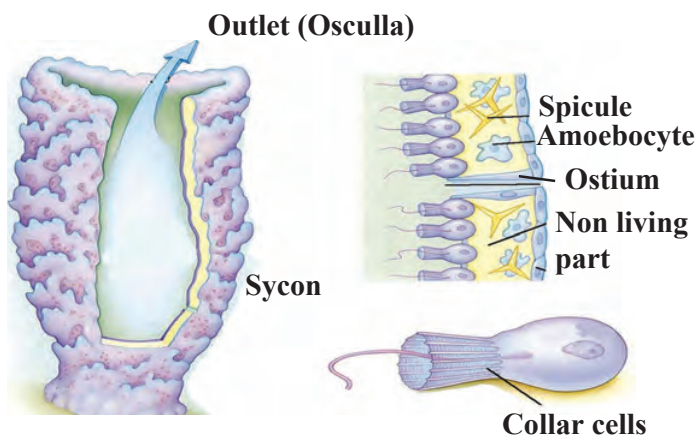


Do you know?

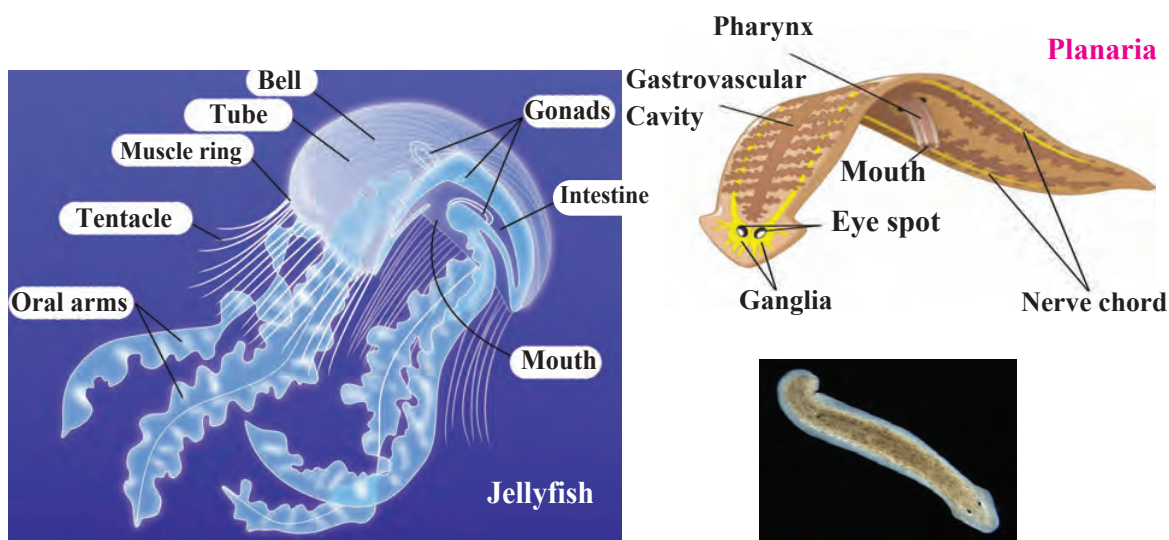
In unicellular organisms all the functions of life are performed by same cell. Body organization of these unicellular organisms is called 'Protoplasmic grade' of organization



6.4 Protoplasmic-grade organization.



6.5 Cellular grade organization



6.6 Cell -Tissue grade organization

6.7 Tissue -Organ grade organization



Observe

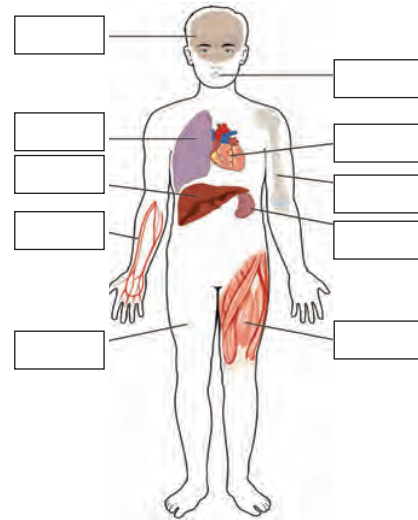
Body organization of human has been shown in the following figure. Use appropriate labels in that figure. Which organs are present in human body?

Besides the four types of body organizations mentioned above, Remaining all animals show 'Organ - system grade organization' in which different organs are joined together to form organ system that performs specific functions. Ex. Crab, Frog, Human, etc.

B. Body Symmetry

Take the pictures of human body and spongilla and try to take an imaginary section through specific plane of their bodies so as to get two equal halves.

What did you observe?



6.8 Organ- System grade of organization

In imaginary sense, if body of any animal is cut through imaginary axis of body, it may or may not produce two equal halves. Depending upon this property, there are different types of animal bodies.

Asymmetrical Body : In case of such body, there is no any such imaginary axis of the body through which we can get two equal halves. Ex. some sponges.

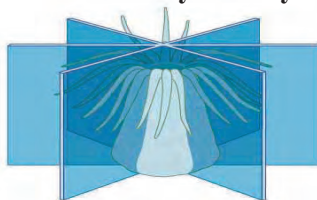
Radial symmetry : In this type of body, if imaginary cut passes through central axis but any plane of body, it gives two equal halves. Ex. Star fish. In case of this animal, there are five different planes passing through central axis of body through which we can get two equal halves.

Bilateral symmetry: In this type of body, there is only one such imaginary axis of body through which we can get two equal halves. Ex. Insects, fishes, frog, birds, human, etc.

Asymmetrical body



Radial Symmetry



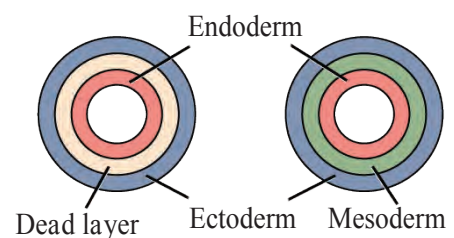
Bilateral Symmetry



6.9 Types of body symmetry

C. Germ Layers: Diploblastic and triploblastic

In case of multicellular animals, germ layers are formed during initial period of their embryonic development and from those germ layers only, different tissues are formed in the body. In case of some animals, only two germ layers [Endoderm & ectoderm] are formed. Ex.: All Cnidarians. In most of all the remaining animals, three germ layers are formed i.e. mesoderm besides endoderm & ectoderm.



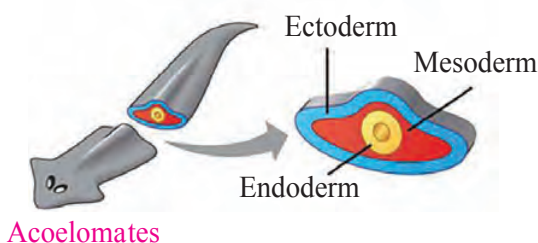
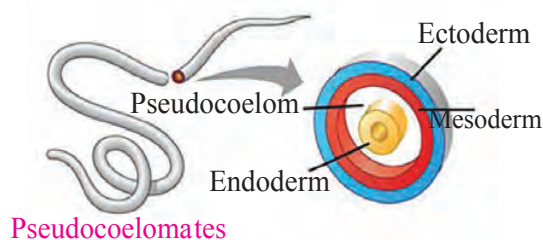
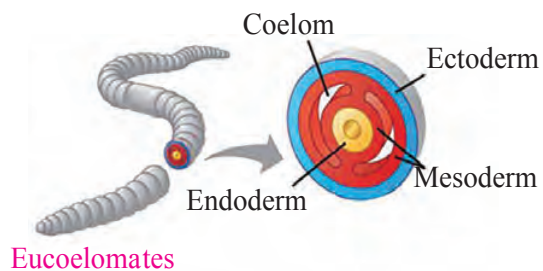
6.10 Diploblastic and triploblastic

D. Body cavity (Coelom)

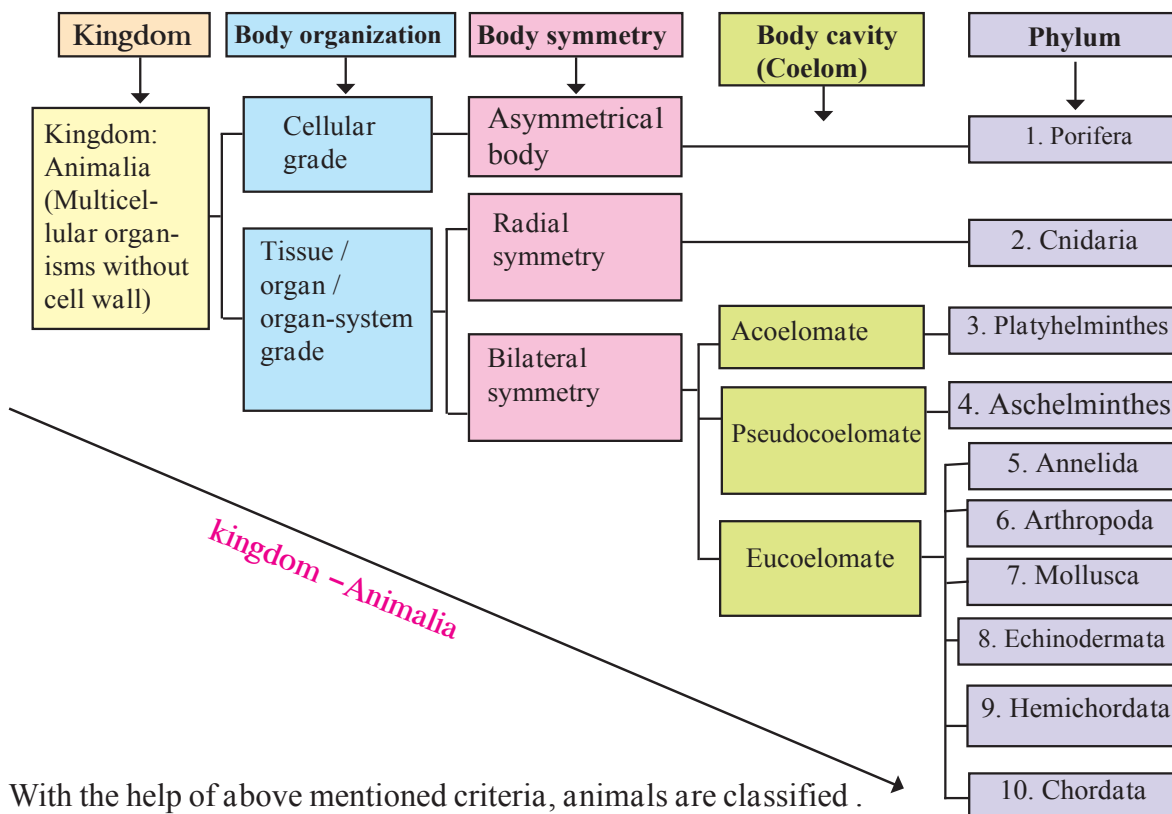
Cavity between the body wall and internal organs is called as body cavity/coelom. In case of multicellular animals, during initial period of their embryonic development, body cavity is formed from either mesoderm. Such type of body cavity is present in animals of phylum Annelida and all phyla coming after Annelida. Such animals are called as eucoelomate (animals with true body cavity). Body cavity is absent in case of animals from phyla Porifera, Cnidaria and Platyhelminthes. Such animals are called as acoelomate. In case of animals from phylum Aschelminthes, they have body cavity but it is not formed by the above mentioned two ways. Hence those animals are called as pseudocoelomates.

E. Body Segmentation

If the body of animals is divided into small, similar units, then such body is called as segmented body and each small unit is called as segment. Ex. Animals like earthworm from phylum Annelida.



6.11 Animal types as per body cavity

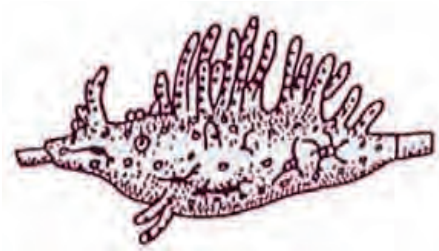


Phylum- Porifera

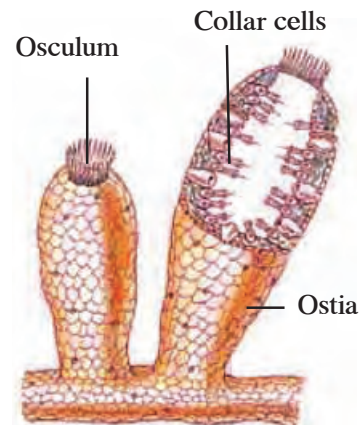
1. These animals are with simplest body plan and are called as 'Sponges'. They bear numerous pores on their body. Those pores are called as 'Ostia' and 'Oscula'.
2. These are aquatic animals. Most of them are marine and few are fresh water dwellers.
3. Most of the animals have asymmetrical body.
4. These animals have special types of cells- collar cells.
5. These animals are always attached to substratum, hence do not show locomotion. Hence, they are referred as sedentary animals.
6. Their spongy body is supported by spicules or spongin fibres. Spicules are made up of calcium carbonate or silica.
7. These animals feed upon small organisms taken in their body along with water. Water is taken in through ostia and given out through oscula.
8. These animals reproduce by budding, an asexual method and / or by sexual method. Besides, they have good ability of regeneration.

Examples: *Sycon*, *Euspongia* (Bath sponge),

Hyalonema, *Euplectella*, etc.



Spongila

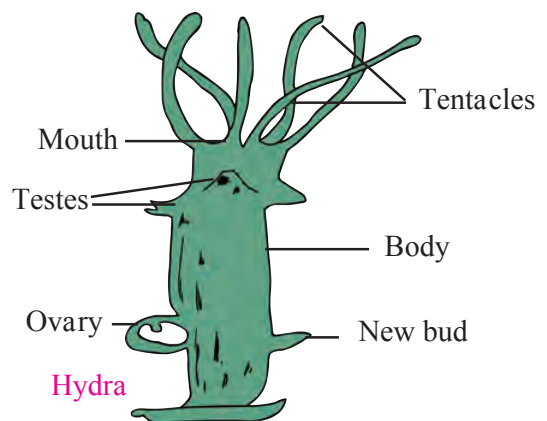


Sycon sponge

6.12 Animals in Porifera phylum

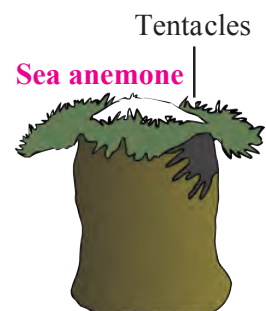
Phylum - Coelenterata/Cnidaria

1. Body of these animals is cylindrical or umbrella-like. If it is cylindrical, it is called as 'Polyp' and if it is umbrella like, it called as 'Medusa'.
 2. Most of these animals are marine. Only few are fresh-water dwellers.
 3. Body of these animals is radially symmetrical & diploblastic.
 4. Cnidoblast bearing tentacles are present around the mouth. Tentacles are useful for capturing the prey whereas cnidoblasts inject the toxin in the body of prey. Those are useful for protection too.
- Examples: *Hydra*, *Adamsia* (Sea anemone), *Physalia* (Portuguese- man-of war), *Aurelia* (Jelly fish), Corals, etc.



Hydra

Coral



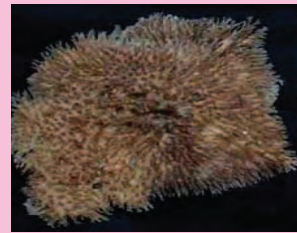
Sea anemone

6.13 Animals in Cnidaria phylum



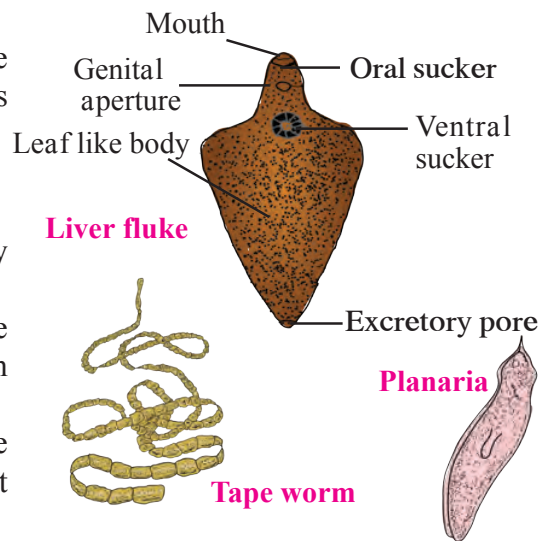
Do you know?

Bath sponge: This is black coloured and somewhat round-shaped animal. Its body is mainly made up of fibers of a protein- spongin and due to this, they have good water-holding capacity. It was used for bathing during old days. Besides, it was also used for manufacturing of pillows and cushions. These were also used as wetting material for sticking postal stamps and counting the currency notes.



Phylum - Platyhelminthes

1. Body of these animals is slender & flat like a leaf or strip. Hence, they are called as 'flatworms'.
 2. Most of these animals are endoparasites. Few are free-living & aquatic.
 3. Body is acoelomate & bilaterally symmetrical.
 4. These are triploblastic i.e. their body is made up of three germ layers- endoderm, ectoderm & mesoderm.
 5. These animals are hermaphrodite i.e. male and female reproductive systems are present in the same animal body.
- Examples: Planaria, Liver fluke, Tape worm, etc.



6.14 Animals in phylum Platyhelminthes

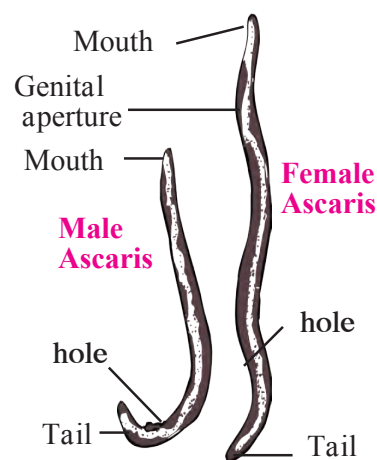
Surprising information

Coral reefs are present in ocean. These reefs are actually colonies of specific cnidarians. A precious stone called 'Coral' (पोवळा) and the coral powder (प्रवाळ भस्म) used in ayurveda is derived from these reefs. Collect more information about these corals from internet.



Phylum- Aschelminthes

1. Body of these animals is long thread-like or cylindrical. Hence, they are called as round worms.
 2. These animals are either free living or endoparasites. Free living animals are either aquatic or terrestrial.
 3. Body of these animals is triploblastic and pseudocoelomate.
 4. Body of these animals is non-segmented and covered with tough cuticle.
 5. These animals are unisexual.
- Examples: *Ascaris* (Intestinal worm), Filarial worm, *Loa loa* (Eye worm), etc.



6.15 Animals in phylum Aschelminthes



Internet is my friend

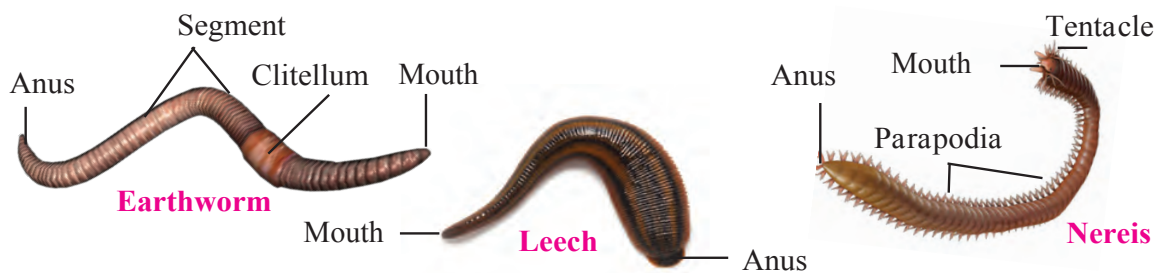
Collect the Information

1. How does the infection of tapeworm in man, liver fluke in grazing animals like goat and sheep occur and what are their preventive measures?
2. How does the infection of round worms like Ascaris, Filarial worm and plant nematodes occur and what are their preventive measures and treatment?

Phylum - Annelida

1. Body of these animals is long, cylindrical and metamerically segmented.
2. Most of the animals are free-living, but few are ectoparasites. Free-living animals may be marine or fresh water dwellers or terrestrial.
3. These animals are triploblastic, bilaterally symmetrical and eucoelomate.
4. They have setae or parapodia or suckers for locomotion.
5. Their body is covered with special cuticle.
6. These animals are either hermaphrodite or unisexual.

Examples: Earthworm, Leech, *Nereis*, etc.



6.16 Animals in phylum Annelida



Get Information

1. Why is earthworm called as friend of farmers?
2. How may be the leech used in ayurvedic system of treatment ?

Phylum- Arthropoda

1. These animals have jointed appendages. Hence they are called as arthropods.
2. Planet Earth has highest number of animals from this phylum. Hence, this is largest phylum with highly successful animals in animal kingdom.
3. These animals are found in all types of habitats ranging from deepest oceans to highest mountains.
4. Body of these animals is triploblastic, eucoelomate, bilaterally symmetrical and segmented.
5. Chitinous exoskeleton is present around their body.
6. These animals are unisexual.

Examples: Crab, spider, scorpion, millipede, centipede, cockroach, butterfly, honey bee, etc.



Find out

What is chitin?

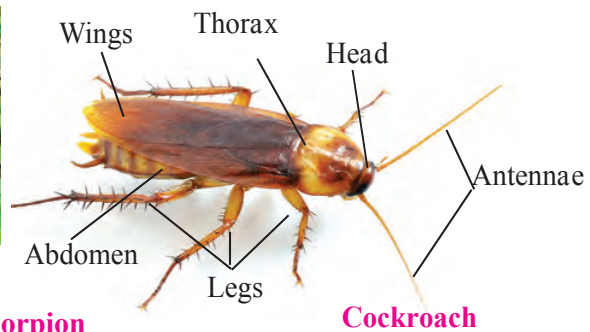
Centipede



Butterfly



Scorpion



Cockroach



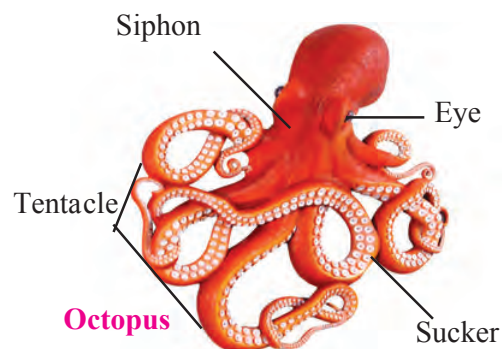
Let's Think

1. What types of benefit & harm occur to human from animals of phylum- Arthropoda?
2. Which are the animals from phylum Arthropoda those have shortest & longest lifespan?
3. Why has it been said that only insects directly compete with humans for food?

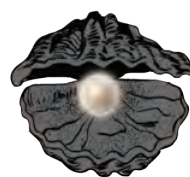
6.17 Animals in phylum Arthropoda

Phylum- Mollusca

1. Body of these animals is soft and slimy. Hence they are referred as mollusc.
2. This is second largest phylum in animal kingdom.
3. These animals are aquatic or terrestrial. Most of the aquatic molluscs are marine, but few are fresh water dwellers too.
4. Body of these animals is triploblastic, eucoelomate, non-segmented and soft. Except animals like snail, their body shows bilateral symmetry. Their body is divided into three divisions like head, foot and visceral mass.
5. Visceral mass is covered with mantle. This mantle secretes a hard, calcareous shell. This shell may be external or internal or even absent in some cases.
6. These animals are unisexual.
Examples: Bivalve, Snail, Octopus, etc.



Octopus



Bivalve



Snail

6.18 Animals in phylum Mollusca

Surprising Information!

1. Octopus is most clever animal among all non-chordates. It can change its colour.
2. It can perform three types of locomotions like swimming, creeping & walking.



Books are my friend

Collect the information about pearl production from bivalves by reading appropriate books.

Phylum- Echinodermata

1. Calcareous spines are present on the body of these animals; hence they are called as echinoderms.
2. These animals are found only in ocean.
3. Their body is triploblastic, eucoelomate. And it is radially symmetrical in adult stage. However, they show bilateral symmetry in larval stage.
4. They perform locomotion with the help of tube-feet. Tube feet are also useful for capturing the prey. Some animals are sedentary.
5. They have skeleton made up of calcareous spines and / or ossicles (plates).
6. These animals have good ability of regeneration.
7. These animals are mostly unisexual.

Examples: Star fish, sea-urchin, brittle star, sea-cucumber, etc.



Star fish



Sea-cucumber



Sea-urchin

6.19 Animals in phylum Echinodermata



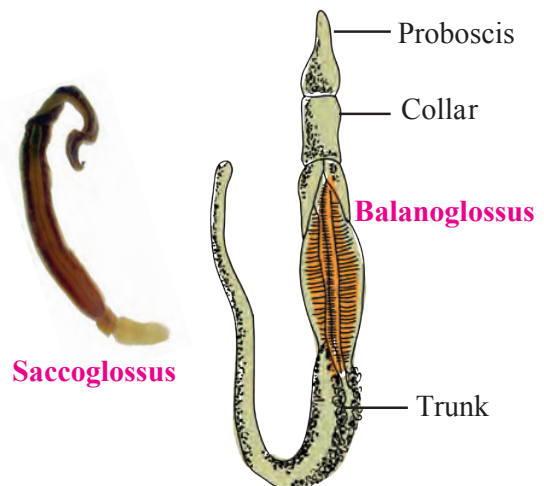
Do you know?

In certain situations, star fish can break apart its body parts and regenerate those later on.

Phylum- Hemichordata

1. Body of these animals is divided into three parts as proboscis, collar & trunk.
2. Notochord is present in proboscis region only. Hence, they are called as hemichordates.
3. These animals are also called as 'acorn worms'.
4. These are marine animals, live in burrows in sand.
5. They have one to many pharyngeal gill slits.
6. They are unisexual or some may be hermaphrodite.

Ex.: *Balanoglossus*, *Saccoglossus*.



6.20 Animals in phylum Hemichordata

Through the view point of evolution, *Balanoglossus* is considered as connecting link between non-chordates and chordates. This animal shows the characters of both the groups.

Phylum- Chordata

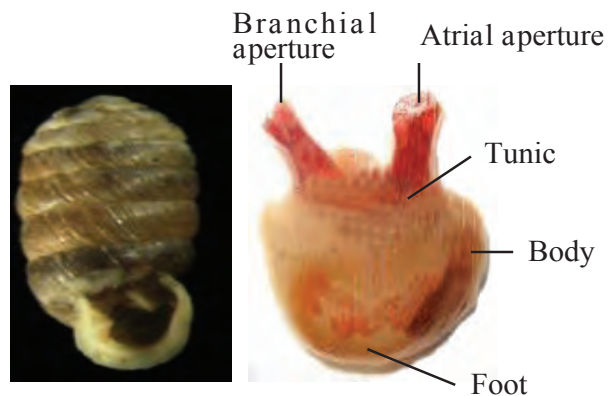
These animals have supporting notochord in their body. All chordates are included in the same phylum. The phylum Chordata is classified in to three subphyla. Following are important characters of phylum Chordata

1. Notochord is present in the body during at least any developmental stage.
2. Pharyngeal gill slits are present in the body during at least any developmental stage.
3. Single, tubular spinal cord is present on dorsal side of body.
4. Heart is present on ventral side of body.

A. Sub phylum - Urochordata

1. These are marine animals.
2. Their body is covered by skin-like test or tunic.
3. Larvae of these animals are freely swimming and notochord is present in only tail region of larvae. Hence, they are called as Urochordata.
4. Larvae metamorphose into adults after settling down at bottom of the sea.
5. Generally, these animals are hermaphrodite.

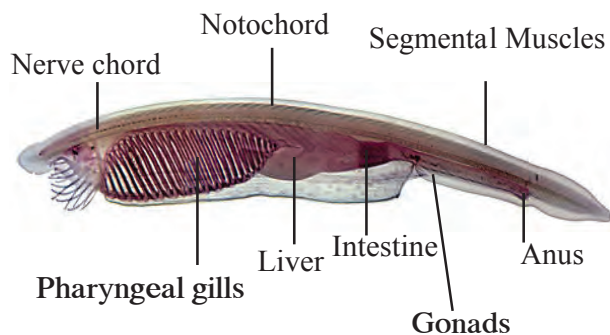
Examples: *Herdmania*, *Doliolum*, *Oikopleura*, etc.



Doliolum

Herdmania

6.21 Animals in Sub- phylum Urochordata



6.21 Animals in Sub- phylum Cephalochordata

B. Sub phylum -Cephalochordata

1. These are small, fish-like, marine animals.
2. Notochord is present throughout the body length.
3. Pharynx is very large and contains gill-slits.
4. These animals are unisexual.
Ex.: *Amphioxus*.

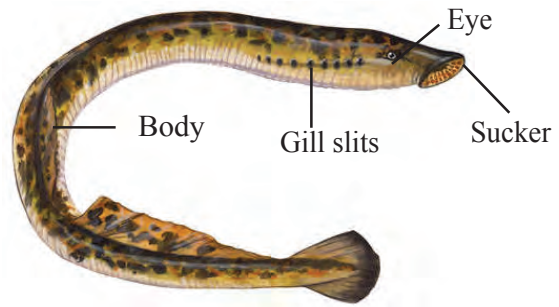
C. Sub phylum -Vertebrata/Craniata

1. In these animals, notochord is replaced by vertebral column.
2. In these animals, head is well developed.
3. Brain is protected by cranium.
4. Endoskeleton is either cartilaginous or bony.
5. Some chordates are jaw-less (Agnatha) whereas some are with jaws (Gnathostomata).

Subphylum- Vertebrata is divided into six classes as follows-

a. Class- Cyclostomata

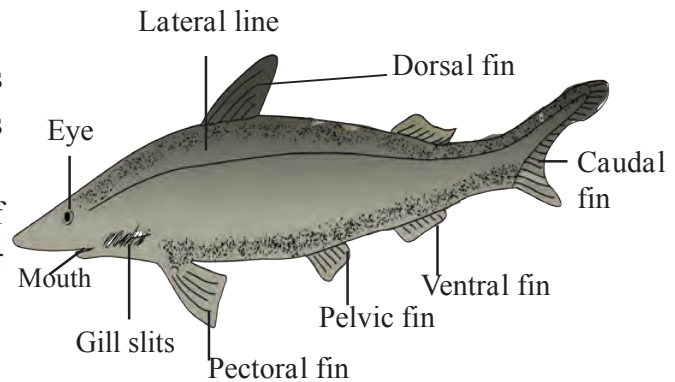
1. These animals have jaw-less mouth provided with sucker.
 2. Their skin is soft and without any scale.
 3. Paired appendages are absent.
 4. Endoskeleton is cartilaginous.
 5. Most of the animals are ectoparasites.
- Examples: Petromyzon, Myxine, etc.



6.23 : Class Cyclostomata-Petromyzon

b. Class- Pisces

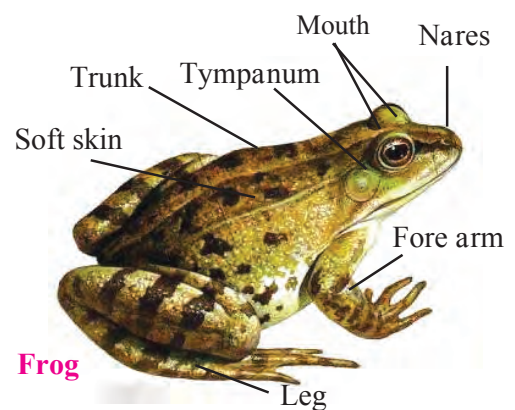
1. These are cold blooded (Poikilotherms) aquatic animals living in marine and fresh waters.
 2. Body is spindle shaped to minimize water-resistance.
 3. They have paired & un-paired fins for swimming. Tail fin is useful as a steering organ during swimming.
 4. Exoskeleton is in the form of scales & endoskeleton is either cartilaginous or bony.
 5. Respiration occurs with gills.
- Examples.: Rohu, Pomfret, Sea horse, Shark, Electric ray, Sting ray, etc.



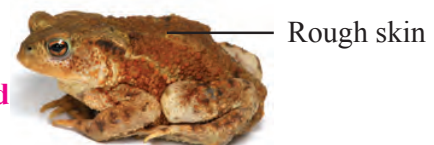
6.24 Class Pisces : Scoliodon (Dog fish)

c. Class- Amphibia

1. These animals are strictly aquatic during larval life and perform only aquatic respiration whereas they can live in water as well as on land during adult life and can perform aquatic as well as aerial respiration.
 2. They have two pairs of appendages. Digits are without claws.
 3. Exoskeleton is absent. Skin is without any derivative and usually kept moist for respiration.
 4. External ear is absent but tympanum is present.
 5. Neck is absent. Eyes are prominent with eye lids.
- Ex.: Frog, Toad, Salamander, etc.



Frog

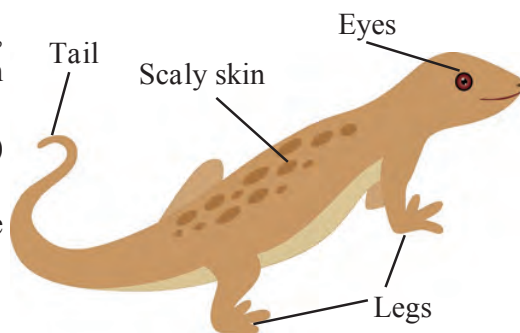


Toad

6.25 Class Amphibia: Frog and Toad

d. Class- Reptilia

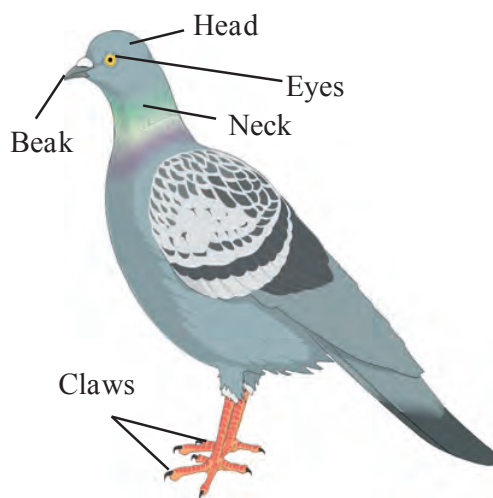
1. According to the course of animal evolution, these are first true terrestrial animals with creeping movement.
2. These are cold blooded (poikilotherms) animals.
3. They creep on the land as their body cannot be lifted up.
4. Their skin is dry and scaly.
5. Neck is present between head & trunk.
6. External ear is absent.
7. Digits are provided with claws.
Examples: Tortoise, Lizard, Snake, etc.



6.26 Class-Reptilia : Wall lizard

e. Class- Aves

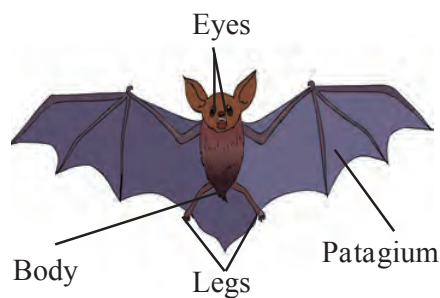
1. These vertebrates are completely adapted for aerial life.
2. These are warm blooded (Homeotherms) i.e. they can maintain their body temperature constant.
3. Their body is spindle-shaped to minimize air resistance during flight.
4. Forelimbs are modified into wings. Digits are covered with scales and bear claws.
5. Exoskeleton is present in the form of feathers.
6. Neck is present between head and trunk.
7. Jaws are modified into beak.
Examples: Peacock, Parrot, Pigeon, Duck, Penguin, etc.



6.27 Class- Aves: Pigeon

f. Class- Mammalia

1. Presence of mammary glands is typical character of mammalia.
2. These animals are warm blooded.
3. Body is divided into head, neck, trunk and tail.
4. Digits are provided with nails, claws, or hooves.
5. Exoskeleton is in the form of hairs or fur.
Examples: Elephant, Human, Kangaroo, Dolphin, Bat, etc.



6.28 Class-Mammalia: Bat



Use your brain power

1. Animals like ghariyal & crocodile live in water as well as on land. Are they amphibians or reptiles?
2. Animals like whale, walrus live in water (ocean). Whether they are included in pisces or mammalia?

Use of Information Technology

Prepare the presentation of animal classification using video clips downloaded from internet.

Books are My Friends

Read the book- The Animal Kingdom: Libbie Hyman and some other similar books.



Always remember

Varieties of animals are found in our surroundings. We should be cautious about not causing any harm to animals during their studies and observations.



Exercise

1. Identify me.

- I am diploblastic & acoelomate. Which phylum do I belong to?
- My body is radially symmetrical. Water vascular system is present in my body. I am referred as fish though I am not. What is my name?
- I live in your small intestine. Pseudocoelom is present in my thread like body. In which phylum will you include me?
- Though I am multicellular, there are no tissues in my body. What is the name of my phylum?

2. Write the characters of each of the following animals with the help of classification chart.

Bath sponge, grasshopper, rohu, penguin, frog, lizard, elephant, jellyfish.

3. Write in brief about progressive changes in animal classification.

4. What is the exact difference between grades of organization and symmetry? Explain with examples.

5. Answer in brief.

- Give scientific classification of shark up to class.
- Write four distinguishing characters of phylum- Echinodermata.
- Distinguish between butterfly and bat with the help of four distinguishing properties.

- To which phylum does Cockroach belong? Justify your answer with scientific reasons.

6. Give scientific reasons.

- Though tortoise lives on land as well as in water, it cannot be included in class- Amphibia.
- Our body irritates if it comes in contact with jelly fish.
- All vertebrates are chordates but all chordates are not vertebrates.
- Balanoglossus is connecting link between non-chordates & chordates.
- Body temperature of reptiles is not constant.

7. Answer the following questions by choosing correct option.

- Which special cells are present in the body of sponges (Porifera)?
1. Collar cells. 2. Cnidoblasts.
3. Germ cells. 4. Ectodermal cells.
- Which of the following animals' body shows bilateral symmetry?
1. Star fish. 2. Jelly fish.
3. Earthworm. 4. Sponge.
- Which of the following animals can regenerate it's broken body part?
1. Cockroach. 2. Frog.
3. Sparrow. 4. Star fish.
- Bat is included in which class?
1. Amphibia. 2. Reptilia.
3. Aves. 4. Mammalia.

8. Complete the following chart.

Body cavity	Germ Layer	Phylum
Absent	-----	Porifera
Absent	Triploblastic	-----
Pseudocoelom	-----	Aschelminthes.
Present	-----	Arthropoda

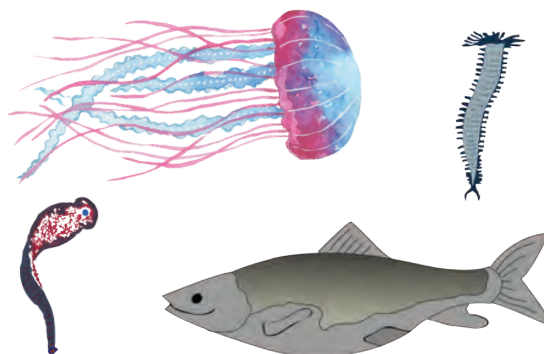
9. Complete the following chart.

Type	Character	Example
Cyclostomata		
	Gill respiration	
Amphibia		
		Whale
	Poikilotherms	

10. Sketch, label and classify

Hydra, Jellyfish, Planaria, Round worm, Butterfly, Earthworm, Octopus, Star fish, Shark, Frog, Wall lizard, Pigeon.

11. Label the following.



Project :

In each week, on a specific day of your convenience, observe the animals present around your school & residence. Perform this activity for six months. Keep datewise record of your observations. After the observation period of six months, analyze your observations with respect to seasons. With the help of your teacher, classify the reported animals.



Photos with courtesy of : Shri Suresh Isave

7. Introduction to Microbiology



- **Applied Microbiology**
- **Industrial Microbiology**
- **Products**



Can you recall?

1. Which different microbes are useful to us?
2. Which different products can be produced with the help of microbes?

Applied Microbiology

Branch of biology in which study of the enzymes related to some prokaryotes and eukaryotic microbes, proteins, applied genetics, molecular biology, etc. is performed; is called as applied microbiology. This study is used for the society and various products like food and medicines are produced on large scale with the help of microorganisms.

Industrial microbiology

This science is related to commercial use of microbes in which various economic, social and environment related processes and products are included. Various microbial processes useful for this purpose are carried out.

Industrial Microbiology: main features

- A. Various productions with the help of fermentation process. Ex. Bread, cheese, wine, raw material for chemicals, enzymes, nutrients, medicines, etc.
- B. Use of microbes for garbage management and pollution control.



Can you recall?

We use the fermentation process while conversion of milk into yoghurt. Which microbes are useful for this process?

Products

A. Dairy Products

Since ancient days, milk is converted into various products for its preservation purpose. Ex. Cheese, butter, cream, kefir, yoghurt, etc. Water content and acidity of the milk changes during formation of these products and texture, taste and flavour is improved.

These processes are performed on large scale with more skill. For production of most of the milk products, bacteria in milk itself are used; only cheese is produced with the help of fungi. Basic process for production of yoghurt, cheese and cream is same. Milk is pasteurized at the beginning to destroy unwanted microbes. It is then fermented with the help of lactobacilli. In this process, lactose sugar of the milk is converted into lactic acid and milk proteins are coagulated with the help of lactic acid. Besides, compounds with taste and flavour are also formed. Ex. Diacetyl has the flavour of butter.

2. Yoghurt Products

Yoghurt is a milk product produced with the help of lactobacilli (inoculant). For maintaining the protein content, condensed milk powder is mixed with milk to be fermented for industrial production of yoghurt. Milk is boiled and once it cools to warm temperature, bacterial strains of *Streptococcus thermophilus* and *Lactobacillus delbrueckii* are added to it in 1:1 proportion. Lactic acid is formed due to *Streptococcus* that makes the proteins to gel out that gives dense consistency to the yoghurt.

Acetaldehyde like compounds are formed due to lactobacilli that gives characteristic taste to the yoghurt. Now a day, various fruit juices are mixed with yoghurt to impart different flavours. Ex. Strawberry yoghurt, banana yoghurt, etc. Shelf life of yoghurt and its probiotic properties can be improved by pasteurization.

3. Butter

Two types of butter like sweet cream and cultured are produced on large scale. Microbes are used for production of cultured variety.

4. Cheese production

Cheese is produced on large scale from the abundantly available cow milk all over the world. First, chemical and microbiological tests of milk are performed. Some colours and microbes like *Lactobacillus lactis*, *Lactobacillus cremoris*, and *Streptococcus thermophilus* are mixed with milk. It imparts sourness to the milk. After this, to impart the dense texture, whey (water in yoghurt) needs to be removed.

An enzyme, rennet obtained from alimentary canal of cattle was being traditionally used earlier. However, an enzyme protease obtained from fungi is used at present to produce vegetarian cheese.

The whey is separated from yoghurt (which has some other uses). Then, process of production of cheese is started through steps like cutting the solid yoghurt into pieces, washing, rubbing, salting, and mixing of essential microbes, pigments and flavours. Then, cheese is pressed and cut in to pieces and stored for ripening.



7.1 Cheese and Butter



Let's think

1. Which different types of cheese are used in western food like pizza, burger, sandwich, etc?
2. What is difference between those types of cheese?



Do you know?

During industrial production of milk products, strict cleanliness and sterilization is essential as bacteria can be attacked by viruses. Hence, virus-resistant varieties of bacteria are developed. Recently, use of mutant varieties of bacteria has been increased. Artificially, some strains are developed that will help to avoid unnecessary steps / materials.

Freshly prepared cheese is always soft e.g. cottage cheese, cream cheese, mozzarella cheese. On storing for 3 to 12 months, semi-hard cheddar cheese is formed whereas after ripening for 12 – 18 months, very hard cheese called parmesan cheese is formed.



Can you tell?

What for probiotic food is famous?

Probiotics

These are also milk products, but contain active bacteria e.g. *Lactobacillus*, *Acidophilus*, *Lactobacillus casei*, *Bifidobacterium bifidum*, etc. These microbes maintain the balance of intestinal microorganisms i.e. increase the population of microbes helping the digestion and decrease the population of harmful microbes (ex. *Clostridium*). Probiotic products are available in various forms like yoghurt, kefir, sauerkraut (pickle of cabbage), dark chocolate, miso soup, pickles, oils, corn syrup, artificial sweeteners, microalgae (Sea food like *Spirulina*, *Chlorella*, Blue green algae, etc.).

Why the probiotics have developed much importance in recent days? These products form the colonies of useful microbes in alimentary canal and control other microbes and their metabolic activities, improve resistance and lower the ill-effects of harmful substances formed during metabolic activities. Useful microbes become inactive due to antibiotics; probiotics make them active again.

Nowadays, probiotics are used for treatment of diarrhoea and treatment of poultry also.



7.2 Probiotics



Use your brain power

In the earlier class, you had prepared the solution of dry yeast for observation of yeast. Which substance is prepared by its use on commercial basis?

Bread

Different types of breads are produced from flour of cereals. Dough is formed by mixing of baker's yeast – *Saccharomyces cerevisiae*, water, salt and other necessary materials with flour. Due to fermentation of carbohydrates by yeast, sugar is converted in to carbon dioxide (CO₂) and ethanol. Dough rises up due to CO₂ and the bread becomes spongy.

Compressed yeast is used in commercial bakery industry. It is available in dry, granular form for domestic use. Yeast produced for commercial use contains various useful contents like carbohydrates, fats, proteins, various vitamins, and minerals. Due to this, bread and other products produced with the help of yeast become nutritive. Ingredients like vinegar, soya sauce and monosodium glutamate (ajinomoto) that are used in presently popular chinese food are produced by microbial fermentation.

Vinegar Production

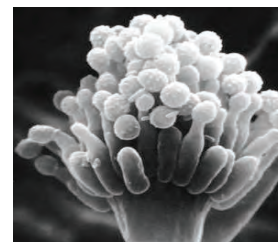
Vinegar is used in each country of the world to impart sour taste to food materials and for preservation of pickles, sauce, ketch-up, chutneys, etc. Chemically, vinegar is 4% acetic acid (CH₃COOH).

Ethanol, an alcohol is obtained by fermentation of carbon compounds like fruit juices, maple syrup, sugar molasses, starch of the roots; with the help of yeast *Saccharomyces cerevisiae*.



7.3 Vinegar

Mixture of bacterial strains like *Acetobacter* and *Glucanobacter* is mixed with ethanol for its microbial degradation. Acetic acid and other by-products are obtained through it. Acetic acid is separated from mixture by rarefaction. Acetic acid is bleached with the help of potassium ferrocynide. Then, it is pasteurized. Finally, very small quantity of SO₂ gas is mixed to produce vinegar.



Soya sauce is produced by fermentation of the mixture of flour of wheat or rice and soyabean with the help of the fungus *Aspergillus oryzae*.

7.4 *Aspergillus oryzae*

Production of beverages

Sr. No.	Fruit	Microbe used	Role of microbe	Name of beverage
1	<i>Coffea arabica</i>	<i>Lactobacillus brevis</i>	Separating seeds from fruit	Coffee
2	<i>Theobroma cacao</i>	<i>Candida</i> , <i>Hansenula</i> , <i>Pichia</i> , <i>Saccharomyces</i> .	Separating seeds from fruit	Cocoa
3	Grapes	<i>Saccharomyces cerevisiae</i>	Fermentation of juice	Wine
4	Apple	<i>Saccharomyces cerevisiae</i>	Fermentation of juice	Cider



Saccharomyces cerevisiae



Fruit and seeds of coffee



Cocoa seeds

7.5 Some ingredients of beverages production



Can you tell?

1. Which functions are performed by enzymes secreted in human digestive system?
2. Give names of some such enzymes.

Microbial Enzymes

Nowadays, instead of chemical catalysts, microbial enzymes are used in chemical industry. These enzymes are active at low temperature, pH and pressure; due to which energy is saved and erosion-proof instruments are also not necessary. Enzymes carry out specific processes; hence unnecessary by-products are not formed due to which expenses on purification are minimised.

In case of microbial enzymatic reactions, elimination and decomposition of waste material is avoided and enzymes can be reused. Hence, such enzymes are eco-friendly. Some examples of microbial enzymes are oxido-reductases, transferases, hydrolases, lyases, isomerases, ligases, etc.

Process of dirt / muck removal occurs at low temperature too due to mixing of enzymes with detergents. Glucose and fructose syrup can be obtained from corn flour by action of enzymes obtained from bacilli and streptomyces. Microbial enzymes are used in various industries like cheese, plant extracts, textile, leather, paper, etc.



Use your brain power

Food materials like cold drinks, ice creams, cakes, juices are available in various colours and flavours. Whether these colours and flavours are really derived from fruits?

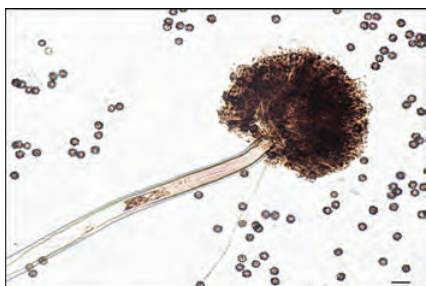


Search

Read the ingredients and their proportion printed on bottles of cold drinks and juices and wrappers of ice creams. Find out the natural and artificial ingredients. Wheat or rice flour is inoculated with *Aspergillus*.

Organic acids used in various commercial products and microbes useful for the same

Source	Microbe	Amino acid	Use
Sugar and beet molasses, ammonia salt	<i>Brevibacterium</i> , <i>Corynebacterium</i>	L-glutamic acid	Production of monosodium glutamate (Ajinomoto)
Sugar molasses, salt	<i>Aspergillus niger</i>	Citric acid	Drinks, toffees, chocolate production
Glucose, corn steep liquor	<i>Aspergillus niger</i>	Gluconic acid	Production of minerals used as supplement for calcium and iron
Molasses, corn steep liquor	<i>Lactobacillus delbrueckii</i>	Lactic acid	Source of nitrogen, production of vitamins.
Molasses, corn steep liquor	<i>Aspergillus itaconius</i>	Itaconic acid	Paper, textile, plastic industry, gum production



7.6 *Aspergillus niger*

What is the xanthan gum that imparts thickness to your favourite ice creams, puddings, chocolates, milk shakes, chocolate drinks, instant soups, etc.? This gum is obtained by fermentation of starch and molasses with the help of *Xanthomonas* species. It is variously useful due to properties like solubility in hot and cold water, high density, etc. It is used for production of pigments, fertilizers, weedicides, textile pigments, tooth pastes, high quality paper, etc.

Substances obtained by microbial processing and their roles

Substances obtained by microbial processing	Roles
Citric, Malic and Lactic acid	To impart acidity
Glutamic acid, Lysine, Tryptophan	Protein binding
Nycin and natamycin	Microbial restrictor
Ascorbic acid (Vit. C), B ₁₂ , B ₂	Antioxidants, vitamins.
Beta carotene, lycopenes, xanthenes, lutein	Edible colours
Polysaccharides, glycolipids	Emulsifiers
Vanillin, Ethyl butyrate (fruit flavour), peppermint flavour, essence of various fruits and flowers	Essence
Xylitol, aspartame	Artificial sweetener (low calorie)



Can you recall?

1. What do you mean by antibiotic?
2. Which precautions should be taken about their consumption?

Antibiotics

Many diseases of human and other animals have been controlled due to antibiotics obtained from different types of bacteria and fungi. Antibiotics like penicillin, cephalosporins, monobactam, bacitracin, erythromycin, gentamycin, neomycin, streptomycin, tetracyclins, vancomycin, etc. are used against various strains of gram positive and gram negative bacteria. Rifamycin is effective against tuberculosis.

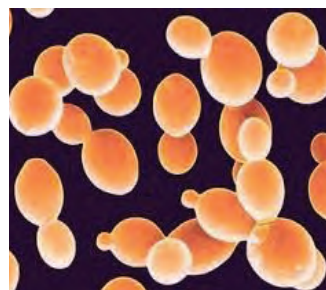


Can you tell?

1. Which different materials are decomposed in biogas plant?
2. Which useful materials are obtained through it? Which is the fuel out of those?
3. Decomposition occurs through which organisms?

Microbes and Fuels

1. Gaseous fuel- methane can be obtained by microbial anaerobic decomposition of urban agricultural and industrial waste.
2. Ethanol, an alcohol is a clean (smokeless) fuel obtained during fermentation of molasses by the yeast- *Saccharomyces*.
3. Hydrogen gas is considered to be the fuel of future. Hydrogen gas is released during bio-photolysis of water in which bacteria perform the photoreduction.



7.7 *Saccharomyces* (yeast)

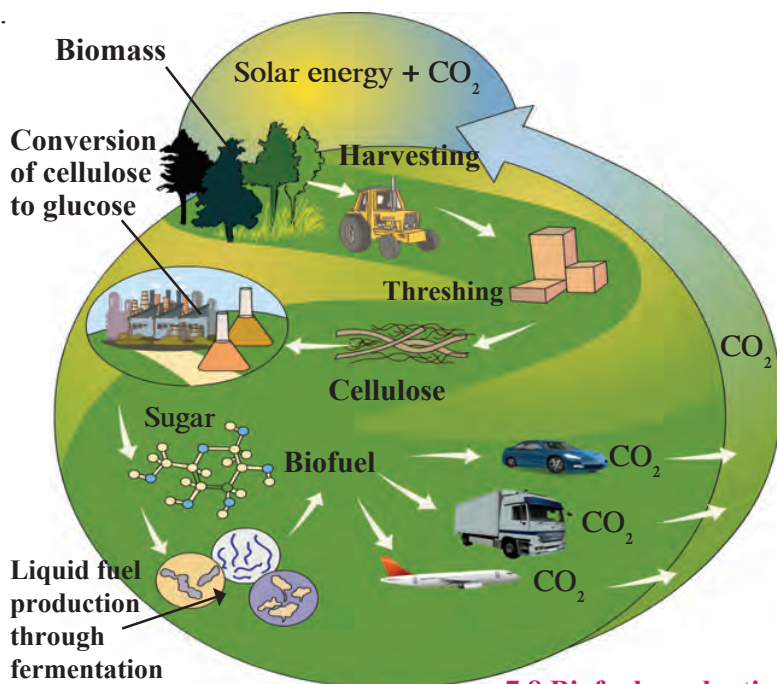
Similar to fuels, various industrial chemicals are also produced through microbial process. Ex. various alcohols, acetone, organic acids, fatty acids, polysaccharides, that are useful as raw materials in chemical industry. Some of these are useful as raw materials for plastic and food products.



Observe

Observe the fig. 7.7. Discuss about bio-fuel.

Bio-fuel : Biofuel is important among the renewable source of good energy. These fuels are available in solid (coal, dung, crop residue), liquid (vegetable oils, alcohol), gaseous (gobar gas, coal gas) forms. These fuels are easily available and in plenty of quantity. These are reliable fuels of the future.



7.8 Biofuel production

Microbial Pollution Control

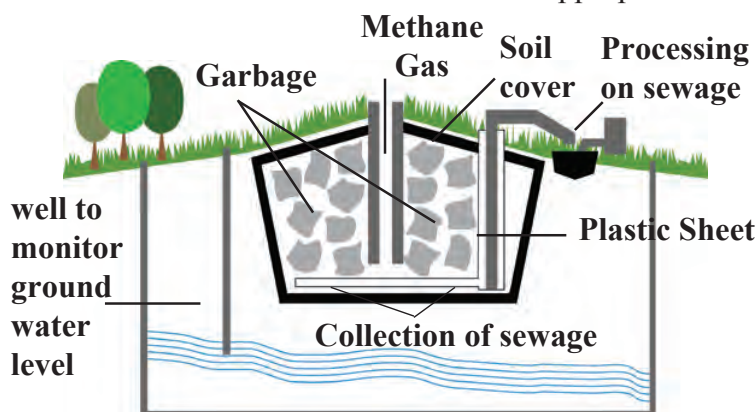
Solids wastes, sewage and various pollutants are ever increasing with increase in pollution. Along with it, ever increasing diseases and degradation of environment are the world-wide problems. Especially, cities in densely populated countries like India are affected with these problems. Life of the future generations will be difficult if these problems are not solved at right time and to right extent. Let us see the role of microbes in environment.

You already know that microbes are used for disposal of solid waste through biogas plant and compost production. How urban waste that is accumulated in terms of tonnes, may be disposed off ?



Let's Think

1. Why is it asked to segregate wet and dry waste in each home?
2. What is done with the segregated waste?
3. Which is most appropriate method of disposal of dry waste?



7.9 Modern landfill site

Land-filling sites

Degradable waste being accumulated in urban areas is used for this purpose. Large pits are dug in open spaces far away from the residential area and those pits are lined with plastic sheets as a precaution against pollution of soil due to leaching of toxic and harmful materials.

Compressed waste is dumped in the pit. It is covered with layers of soil, saw dust, leafy waste and specific biochemicals. Bioreactors are mixed at some places. Microbes present in soil and other top layers decompose the waste. Completely filled pit is sealed with soil slurry. Best quality compost is formed after few days. Such land filling sites can be reused after removal of compost.



Observe

Observe the garbage vans of grampanchayat and municipality. Nowadays, there is facility of decreasing the volume of garbage by compaction in those vans. Explain the advantages of this activity.

Sewage Management

In villages, domestic sewage is disposed off either in nearby soil or in biogas plant. However, in cities, sewage needs to be carried to processing unit and acted upon by microbial processes.

Microbes which can decompose any compound as well as destroy the pathogens of cholera, typhoid, etc. are mixed with sewage. They release methane and CO_2 by decomposition of the carbon compounds present in sewage. Phenol oxidizing bacteria decompose the xenobiotic chemicals present in sewage.

The sludge that settles down in this process can again be used as fertilizer. Water released after microbial treatment is environmentally safe. Microbes are used for bioremediation of environment polluted due to sewage.



Collect Information

1. Which materials should not be present in garbage for its proper microbial decomposition?
2. How the sewage generated in your house or apartment is disposed off?

Clean Technology

Human being has made a very fast progress in technology. However, environmental pollution is also increasing with same speed. Let us see the ways of control over air, soil and water pollution with the help of microbes.

Microbes have natural ability of decomposing the manmade chemicals. Hydrocarbons and other chemicals are transformed with the help of these abilities.

1. Some microbes remove the sulphur from fuels.
2. Metals like copper, iron, uranium, zinc, etc. leach into environment from low quality metalloids. These are converted into compounds before leaching, with the help of thiobacilli and sulphobacilli.



Can you tell?

You must have seen or read the news of dead fishes or oily water accumulating at the sea coasts. Why does this happen?

Spilling of petroleum oil occurs in ocean due to various reasons. This oil may prove fatal and toxic to aquatic organisms. It is not easy to remove the oil layer from surface of water by mechanical method. However, bacteria like *Pseudomonas* spp. and *Alcanovorax borkumensis* have the ability to destroy the pyridines and other chemicals. Hence, these bacteria are used to clear the oil spills. These are called as hydrocarbonoclastic bacteria (HCB). HCB decompose the hydrocarbons and bring about the reaction of carbon with oxygen. CO_2 and water is formed in this process.

Plastic bottles are formed from the chemical substance PET (Polyethylene Terephthalate Polyester). Now a day, most of the urban garbage consists of plastic. It has been observed that species like *Vibrio*, *Ideonella sakaiensis* can decompose the PET. Similarly, species of bacteria like *Actinomycetes*, *Streptomyces*, *Nocardia*, *Actinoplanes* have ability of decomposing rubber from garbage.



7.10 *Alcanovorax borkumensis*



7.11 *Pseudomonas*



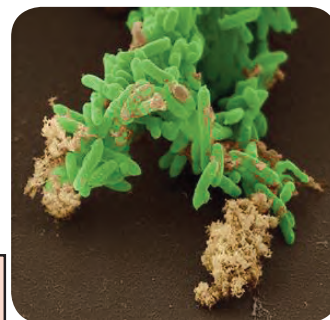
7.12 *Acidobacillus*

Sulphuric acid is present in the acid rain and materials coming out of mines. You know that erosion of metals present in statues, bridges and buildings occurs due to it. Sulphuric acid is source of energy for some species of bacteria like *Acidophillium* spp. and *Acidobacillus ferrooxidans*. Hence, these bacteria can control the soil pollution occurring due to acid rain.



Internet is My Friend

Collect pictures of various useful microbes. Display chart of their information in classroom.



7.13 *Geobacter*

Water soluble salts of uranium are present in the wastes produced during electroplating and in effluent released in environment from the atomic energy plant. *Geobacter* convert these salts of uranium into insoluble salts and thereby prevent those salts from mixing with ground water sources.

Microbes and Farming



Can you tell?

How the bacteria present in soil and root nodules of leguminous plants are useful?

Microbial Inoculants

Some microbes-containing inoculants are produced by process of fermentation. These inoculants are sprayed on seeds before sowing and some of the inoculants are released into plants. Microbes in the inoculants help in plant growth by supplying nutrients. They improve the quality of vegetarian food. Solution containing *Azotobacter* and artificial nitrogenase is used in organic farming.

Soil pollution occurring due to chemical fertilizers is prevented due to use of these solutions. Fluoroacetamide-like chemicals are mixed with soil due to use of chemical pesticides in agriculture. These prove to be harmful to other plants and animals as well as cause skin diseases to human. These pesticides in the soil can be destroyed with the help of microbes.

Bioinsecticides

Bacterial and fungal toxins which can destroy pests and pathogens can be directly integrated into plants with the help of biotechnology. Being toxic to insects, they do not consume the plants. Similar to bacteria, some species of fungi and viruses are useful as pesticides. Spinosad, a by-product of fermentation is a biopesticide.



Always Remember

Plastic being used for storing the garbage is biodegradable polylactic acid. Such material should be used as per need only and environment should be saved.



7.14 Caterpillar feeding on leaf

Exercise

1. Rewrite the following statements using correct of the options and explain the completed statements.

(gluconic acid, coagulation, amino acid, acetic acid, clostridium, lactobacilli)

- Process of -- -- -- of milk proteins occurs due to lactic acid.
- Harmful bacteria in like -- -- -- in the intestine are destroyed due to probiotics.
- Chemically, vinegar is -- -- --.
- Salts which can be used as supplement of calcium and iron are obtained from ----- acid.

2. Match the pairs

'A' group

- Xylitol
- Citric acid
- Lycopene
- Nycin

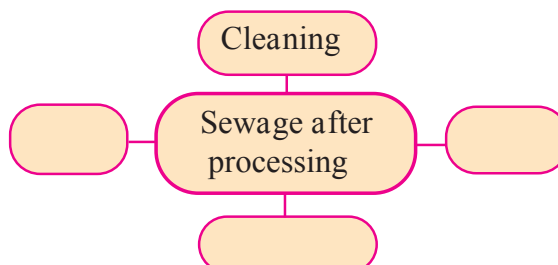
'B' group

- Pigment
- To impart sweetness
- Microbial restrictor
- Protein binding emulsifier
- To impart acidity

3. Answer the following.

- Which fuels can be obtained by microbial processes? Why is it necessary to increase the use of such fuels?
- How can the oil spills of rivers and oceans be cleaned?
- How can the soil polluted by acid rain be made fertile again?
- Explain the importance of bio-pesticides in organic farming.
- Which are the reasons for increasing the popularity of probiotic products?
- How the bread and other products produced using baker's yeast are nutritious?
- Which precautions are necessary for proper decomposition of domestic waste?
- Why is it necessary to ban the use of plastic bags?

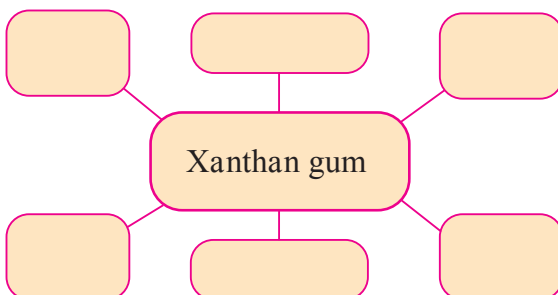
4. Complete the following conceptual picture.



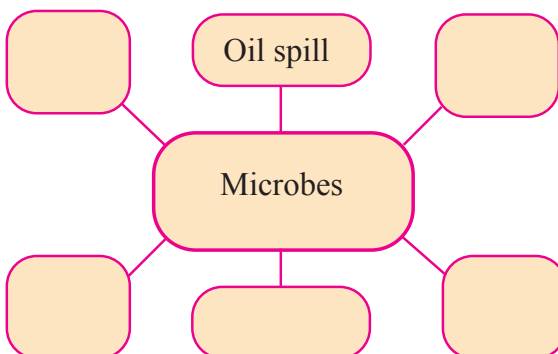
5. Give scientific reasons.

- Use of mutant strains has been increased in industrial microbiology.
- Enzymes obtained by microbial process are mixed with detergents.
- Microbial enzymes are used instead of chemical catalysts in chemical industry.

6. Complete the following conceptual picture with respect to uses.



7. Complete the following conceptual picture related to environmental management.



8. Answer the following.

- a. What is role of microbes in compost production?
- b. What are the benefits of mixing ethanol with petrol and diesel?
- c. Which plants are cultivated to obtain the fuel?
- d. Which fuels are obtained from biomass?
- e. How does the bread become spongy?

Project :

1. Find the ways to implement the zero garbage system at domestic level.
2. Which are the microbes that destroy the chemical pesticides in soil?
3. Collect more information about reasons for avoiding the use of chemical pesticides.



8. Cell Biology and Biotechnology



➤ **Cell Biology (Cytology)**

➤ **Biotechnology and Its Applications**

➤ **Stem Cells**

➤ **Important Stages in Development of Agriculture**



Can you Recall

1. What is cell?
2. What is tissue? Which are the functions of tissue?
3. Which technique in relation to tissues have you studied in earlier classes?
4. Which are the various processes in tissue culture?

We have studied the plant production by technique of tissue culture in the last class. Stem cells are used for that purpose. Whether such stem cells are present in animals?



Observe

Assign names in the figure given below. Explain the various stages those are kept blank.

Cytology

Earlier, we have studied the structure, types and organelles of the cell. This is called as cell biology / cytology. Besides above mentioned points, it includes the study of cell division and many other aspects of the cell.

There are revolutionary changes in the field of human health due to cell biology. Research institutes specially dedicated for research on cells are established at Pune and Bengaluru, India. National Center for Cell Science (<http://www.nccs.res.in>) at Pune and 'Instem' (<http://www.instem.res.in>) at Bengaluru are involved in valuable research.

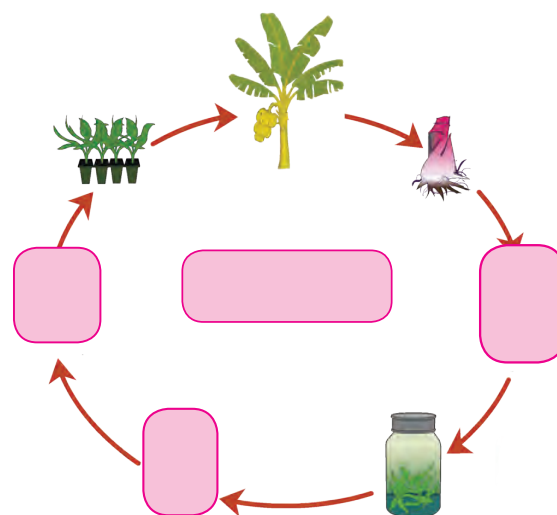
Visit both the websites mentioned above and with the help of your teacher, try to understand the research undergoing in those institutes.

Stem Cells

These are special types of cells present in the body of multicellular organisms. These cells give rise to all other types of cells present in the body of multicellular organisms. Similarly, these cells play an important role in wound healing.

We had studied the stem cells of plants in the previous class. Now, we shall study the stem cells in animals, particularly in human beings.

New organism is formed from the zygote that is formed by union of male and female gamete. At the earliest stage of development, organism is in the form of a mass of cells. All the cells in that mass are almost alike. Those cells are called as stem cells.



8.1 Diagram

During further development, these cells form any type of cell, different types of tissues and perform different functions in the body. This is the differentiation of stem cells. However, once the tissues are formed, the cells in those tissues, at the most, can form same types of cells only. This is the case in each part of the body. However, stem cells are present for longer duration in some parts of the body.

Stem cells are present in the umbilical cord by which the foetus is joined to the uterus of the mother. Stem cells are also present in the blastocyst stage of embryonic development. Stem cells are present in red bone marrow and adipose connective tissue of adult human beings. It has become possible to produce different types of tissues and the degenerated part of any organ with the help of these stem cells.

Stem Cell Preservation

For the purpose of preservation, stem cell samples are carefully collected from sources like cord blood, red bone marrow or embryo (blastocyst) and are kept in small, sterile vials. Those vials are kept in liquid nitrogen at -135°C to -190°C .



Use your brain

Just like the grafting in plants, is the organ transplantation possible in humans?

Stem cell research

In biotechnology, stem cell research is a revolutionary event after cloning. This technique has the potential of bringing about the fundamental changes in the medical science.

Depending upon source, stem cells are of two types as embryonic stem cells and adult stem cells.

Embryonic stem cells

Division of the zygote starts and thereby it is converted into embryo. Cells of embryo undergo repeated mitotic divisions. Cell differentiation starts from 14th day of conception. Cells of different organs like osteocytes (bone cells), hepatocytes (liver cells), and neurons are formed due to differentiation. Embryonic cells before differentiation are called as embryonic stem cells. 220 different types of cells in human body are formed from single type cells i.e. embryonic stem cells. Thus, stem cells are primary type of undifferentiated cells with self-multiplying ability and they are parent cells of all types of human cells. This property of stem cells is called as pluripotency. It has been found that if these stem cells are collected well before the beginning of differentiation on 14th day i.e. during 5th – 7th day and cultured with certain biochemical stimulus in laboratory, as per the stimulus, they can transform themselves into desired type of cells, thereby tissues and finally into organs.

Adult stem cells

Stem cells can be obtained from the body of adult person too. There are three main sources of stem cells in the body of adult persons. Stem cells can be obtained from red bone marrow, adipose connective tissue and blood. Besides, stem cells can be obtained from cord blood immediate after birth.

Uses of Stem Cells

1. Regenerative Therapy

A. Cell Therapy: Stem cells are used to replace the dead cells in case of conditions like diabetes, myocardial infarction, Alzheimer's disease, Parkinson's disease, etc.

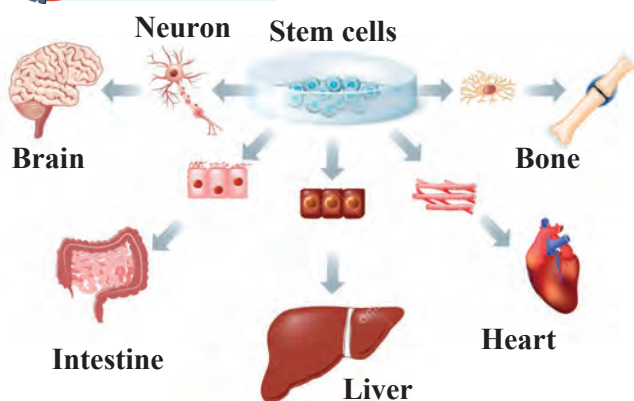
B. To produce blood cells required in conditions like anaemia, thalassemia, leukemia, etc.

2. Organ Transplantation: In case of failure of organs like kidney and liver, those can be produced with the help of stem cells and transplanted.

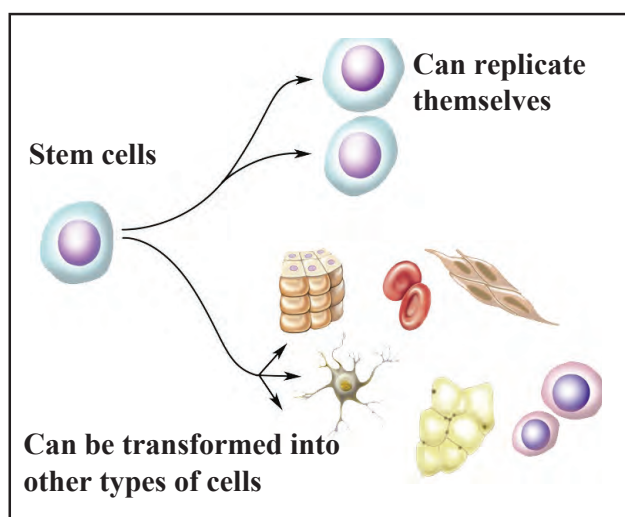


Observe

Discuss about stem cells and organ transplantation in the class with the help of figures given below.



8.2 Stem cells and organs



8.3 Stem cells therapy

Organ transplantation

Various organs in the human body either become less efficient or completely functionless due to various reasons like aging, accidents, infections, disorders, etc. Life of such person becomes difficult or even fatality may occur under such conditions. However, if a person gets the necessary organ under such conditions, its life can be saved.

Availability of donor is an important requirement in organ transplantation. Each person has a pair of kidneys. As the process of excretion can occur with the help of single kidney, person can donate another one. Similarly, skin from certain parts of the body can also be donated.

Various factors like blood group, diseases, disorders, age, etc. of the donor and recipient need to be paid attention during transplantation.

However, other organs cannot be donated during life time. Organs like liver, heart, eyes can be donated after death only. This has led to the emergence of concepts like posthumous (after death) donation of body and organs.

Organ and Body Donation: Human bodies are disposed off after death as per traditional customs. However due to progress in science, it has been realized that many organs remain functional for certain period even after death occurs under specific conditions. Concepts like organ donation and body donation have emerged recently after realization that such organs can be used to save the life of other needful persons. A liberal view behind the concept of organ and body donation is that after death, our body should be useful to other needful persons so that their miserable life would become comfortable. Awareness about these concepts is increasing in our country and people are voluntarily donating their bodies.

Life of many people can be saved by organ and body donation. Blinds can regain the vision. Life of many people can be rendered comfortable by donation of organs like liver, kidneys, heart, heart valves, skin, etc. Similarly, body can be made available for research in medical studies. Many government and social organizations are working towards increasing the awareness about body donation.



Internet is my friend

Visit the websites- <http://www.who.int/transplantation/organ/en/> & www.organindia.org/approaching-the-transplant/ and collect more information about 'brain dead', organ donation and body donation.



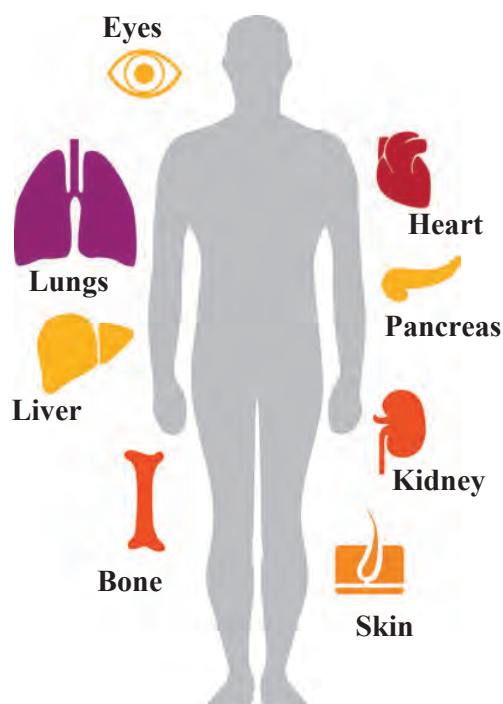
Always Remember

Organ donation and transplantation is under the control of 'Transplantation of Human Organs Act, 1994' and subsequent amendments of 2009, 2011 and 2014 so that overall process would be transparent and any person would not be cheated.



Recall A Little

1. What is biotechnology?
2. In which various fields, the biotechnology has been useful?
3. What is the impact of biotechnology on agriculture and other related fields?



Biotechnology

We have studied in the earlier class that biotechnology is bringing about artificial genetic changes and hybridization in organisms for human welfare. Various branches of science like cytology, biochemistry, molecular biology, and genetic engineering are included in biotechnology. There is considerable progress mainly in the field of agriculture and pharmacy due to biotechnology. New experiments are being performed for improving the agricultural yield. In pharmacy, experiments for production of antibodies, vitamins, and hormones like insulin have been successful. High-class varieties of crops have been developed through the technique of tissue-culture.

Biotechnology includes following main areas

1. Use of various abilities of microbes like yoghurt production from milk and alcohol from molasses.
2. Use of productivity of the cells. Ex. – Production of antibiotics and vaccines, etc. with the help of specific cells.
3. Use of bio-molecules like DNA and proteins in human welfare.
4. Development of plants, animals and products of desired quality by gene manipulation. Production of human growth hormone with the help of genetically modified bacteria.
5. Use of genetic and non-genetic technique. Non-genetic biotechnology involves use of either cell or tissue. Ex. Tissue culture, production of hybrid seeds, etc.

Benefits of Biotechnology

1. It has become possible to increase the per hectare yield irrespective of the limitations of crop-land area.
2. Expenses on disease control have minimized since development of resistant varieties.
3. Due to development of fast fruit setting varieties, yield per annum has been increased.
4. Development of stress resistant varieties which can withstand variable temperature, water-stress, changing fertility of soil, etc. has become possible.

8.4 Organs that can be donated

Development of Biotechnology in India

Government of India had established the National Biotechnology Board in 1982. This board was transformed into department of biotechnology under the ministry of science and technology, in 1986. Various institutes in India are working under the control of this department of biotechnology. It includes National Institute of Immunology, National Facility for Animal Tissue and Cell Culture, National Centre for Cell Science, National Brain Research Centre, Central Institute of Medicinal and Aromatic Plants. There are facilities of higher education and research in these institutes from where thousands of students have pursued Ph.D. degrees and are contributing to the progress of country in the field of biotechnology.

Commercial Applications of Biotechnology:

1. Crop Biotechnology: Biotechnology is used in agricultural field to improve yield and variety.

a. Hybrid Seeds: Genes of two different crops are recombined to form hybrids of various crops. This is especially useful for fruits.

b. Genetically Modified Crops: Crops developed with desired characters by integrating foreign gene with their genome are called as genetically modified crops. High yielding varieties with resistance to diseases, alkalinity, weeds, other stresses like cold and drought.

Bt Cotton: A gene had been isolated from the bacterium *Bacillus thuringiensis* and integrated with the genome of cotton. Due to this, the toxin which is fatal for bollworm was produced in leaves and bolls of cotton. If bollworm feeds on leaves, the toxin destroys its alimentary canal and the bollworm dies.

Bt Brinjal: BT Brinjal variety is developed by using the gene isolated from *Bacillus thuringiensis*. This improved variety of brinjal kills the pest in same way as the Bt cotton does.

Golden Rice: A gene synthesizing the vitamin A (Beta carotene) has been introduced in this variety of rice. As compared to the normal variety, this variety which has been developed in 2005 contains 23 times more amount of beta carotene.

Herbicide tolerant plants: Weeds always affect the growth of main crop. If herbicides are used to destroy the weeds, it affects the main crop too. Due to this, Herbicide tolerant plants varieties of crops are being developed. Due to this, it has become possible to selectively destroy the weeds.



8.5 Some crops

c. Biofertilizers

Due to use of biofertilizers instead of chemical fertilizers, nitrogen fixation and phosphate solubilization abilities of the plants are improved. Mainly the bacteria like *Rhizobium*, *Azotobacter*, *Nostoc*, *Anabaena* and plants like *Azolla* are used as biofertilizers.

In the last year, we have studied the tissue culture. Genetic improvement of the plants has become possible due to tissue culture and besides, those characters inherited to next generation.



8.6 Azolla



Make a list and discuss

Give five examples of each of the fruiting and flowering plants developed through tissue culture and mention their benefits.

2. Animal Husbandry

Two main methods as artificial insemination and embryo transfer are used in animal husbandry. It helps to improve both, the quantity and quality of animal products. Ex. Milk, meat, wool, etc. Similarly, animals with more strength have been developed for hard work.

USE OF ICT

Collect information about various hybrid varieties of animals. What are their benefits? Make a presentation of various pictures and videos.

3. Human Health

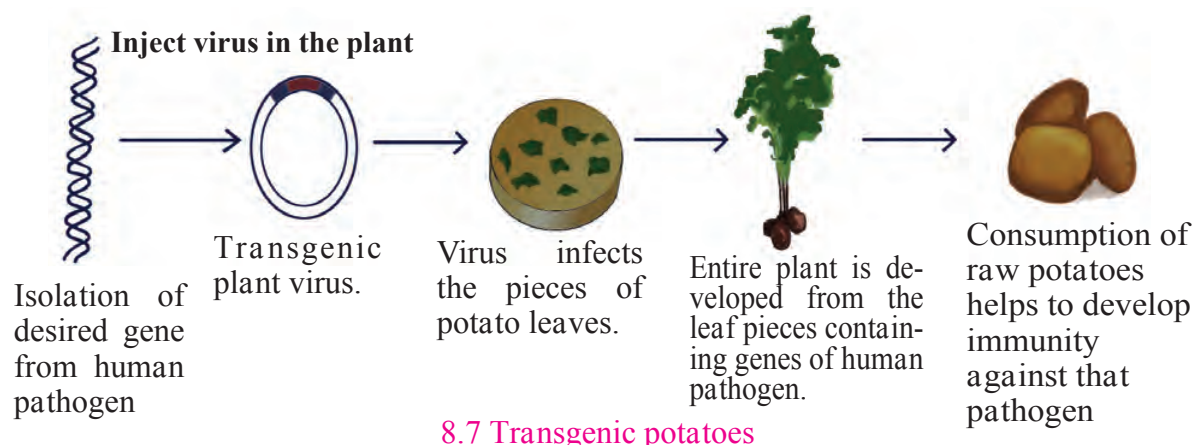
Diagnosis and treatment of the diseases are two important aspects of the human health management. Biotechnology helps to identify the role of gene, if any, in disease of a person. Diagnosis of diabetes and heart diseases has become possible even before the onset of symptoms, with the help of biotechnology. Diagnosis of the diseases like AIDS, dengue can be done within few minutes. Hence, treatment can be done at the earliest.

Various medicines are used for the treatment of diseases. Ex. The hormone insulin is used in treatment of diabetes. Earlier, insulin was being collected from the pancreas of horses. However, nowadays, due to biotechnology, insulin can be prepared with the help of bacteria. For this purpose, human insulin gene has been inserted into the genome of bacteria. Various vaccines and antibiotics are also produced in the same way.

a. Vaccines and Vaccination: Vaccine is the ‘antigen’ containing material given to acquire either permanent or temporary immunity against a specific pathogen or disease. Traditionally, vaccines were prepared with the help of pathogens. Completely or partially killed pathogens were used as vaccines. However, due to this, there were chances of contracting the disease in case of some persons. Hence, as an alternative, scientists tried to artificially produce vaccines with the help of biotechnology. For this purpose, scientists produced the antigen in laboratory with the help of gene isolated from the pathogen and used it as vaccine. Thus, safer vaccines are being produced.

Now, proteins which act as antigen are injected in pure form instead of injecting the killed or semi-killed pathogens. These proteins keep the persons away from the diseases by keeping the immune system active. Thus, injecting the antigens is safest way in vaccination. Vaccines produced with the help of biotechnology are more thermo-stable and remain active for longer duration. Ex. Vaccines of polio, hepatitis,

Edible Vaccines: Work on production of edible vaccines is in progress and presently, potatoes are being produced with the help of biotechnology. These potatoes are called as transgenic potatoes. These potatoes will act against bacteria like *Vibrio cholerae*, *Escherichia coli*. Consumption of these raw potatoes generates the immunity against cholera and the disease caused due to *E. coli*. What will happen if these potatoes are cooked for consumption?



b. Treatment: Biotechnology is useful for production of hormones like insulin, somatotropin and blood clotting factors.

c. Interferon: This is a group of small sized protein molecule used in treatment of viral diseases. These are produced in blood. However, nowadays, with the help of biotechnology, transgenic *E. coli* are used for production of interferon.

d. Gene therapy: Gene therapy to treat genetic disorders in somatic cells has become possible due to biotechnology. Ex. Phenylketonuria (PKU) arises due to genetic changes in hepatocytes (liver cells). It has become possible to treat it with gene therapy. This method is called as somatic cell gene therapy. All the cells except sperms and ova in the body are called as somatic cells



Do you know ?

Indian Institute of Science has developed a transgenic variety of tobacco. If cattles feed upon leaves of this plant, they do not contract a viral disease- rinderpest.

e. Cloning: Production of replica of any cell or organ or entire organism is called cloning.

i. Reproductive cloning: A clone can be produced by fusion of a nucleus of somatic cell with the enucleated ovum of anybody. Thus, there is no need of sperm to produce the new organism.



ii. Therapeutic cloning

Stem cells can be derived from the cell formed in laboratory by the union of somatic cell nucleus with the enucleated egg cell. Various diseases can be treated with the help of these stem cells.

- * Similar to cells, genes can also be cloned and millions of copies of same gene can be produced. Those can be used for gene therapy and other purposes.
- * Controlling the inheritance of hereditary diseases, continuation of generations, enhancing the specific tendency may become possible due to cloning technique. However, there is world-wide opposition to human cloning on various issues.

4. Industrial Products / White Biotechnology

Various industrial chemicals can be produced through less expensive processes. Example: Alcohol production from sugar molasses with the help of transgenic yeast.

5. Environment and Biotechnology

It has become possible to solve environment related various problems with the help of biotechnology.

Microbial techniques are already in use for treatment on sewage and solid waste. Sewage is rich in organic matter. If such sewage is released in natural water bodies like rivers, the organic matter in it gets oxidized with the help of dissolved oxygen. Due to this, level of dissolved oxygen in water decreases, adversely affecting the aquatic life. As a remedy on this, sewage should be released in to rivers only after oxidation with the help of microbial technique.

See the proteins produced by biotechnology and the names of diseases they are used against-

1. Insulin	Diabetes
2. Somatostatin	Dwarfism
3. Erythropoietin	Anaemia
4. Factor VIII	Haemophilia
5. Interleukin	Cancer
6. Interferon	Viral infection

Peeking into History

A sheep 'Dolly' was born in Scotland by cloning technique on 5th July 1996. Nucleus from the udder cell of sheep of 'Finn Dorset' variety had been introduced into enucleated ovum of Scottish sheep. Then, the ovum was allowed to develop in the uterus of Scottish sheep and thereby the 'Dolly' had been born. It was showing the characters as per the chromosomes in nucleus and any character of Scottish sheep was not visible.



Dolly (Clone)

- i. Microbes are useful on large scale while production of compost by treatment on solid organic waste material.
- ii. Bio-remediation, biopesticides, biofertilizers, biosensors, etc, are some new concepts in biotechnological methods.

Bioremediation means either absorption or destruction of toxic chemicals and harmful pollutants with the help of plants and microorganisms. If plants are used for this purpose, it is called as 'phyto-remediation'. Some examples of bioremediation are as follows-

- * The *Pseudomonas* bacteria are useful for cleaning the hydrocarbon and oil pollutants from soil and water.
 - * The fern *Pteris vitata* can absorb the arsenic from the soil.
 - * Genetically modified variety of Indian mustard can absorb selenium from soil.
 - * Sunflower can absorb uranium and arsenic.
 - * The bacterium *Deinococcus radiodurans* is highly radiation resistant organism. It has been genetically modified and used to absorb the radiations from radioactive debris.
 - * Grasses like alfalfa, clover and rye are used in phyto-remediation.
5. **Food Biotechnology:** Food items like bread, cheese, wine, beer, yoghurt, vinegar are produced with the help of microorganisms. These food items are probably the oldest ones produced with the help of biotechnology.
6. **DNA fingerprinting:** DNA sequence of each person is unique as that of the fingerprints. Due to this, identity of any person can be established with the help of its available DNA. This is called as DNA fingerprinting. It is mainly useful in forensic sciences. Identity of the criminal can be established with the help of any part of its body found at the site of crime. Similarly, identity of father of any child can be established. This research is performed in Center for DNA fingerprinting and Diagnostics, Hyderabad.

Cleaning of Oil Spillage in Oceans: If oil spillage occurs, it adversely affects the marine life. Now, cleaning the ocean without any harm to environment in cheaper way has become possible with the help of oil-digesting and fast multiplying bacteria. India born American citizen and scientist Dr. Anand Mohan Chakravarti had for the first time suggested the use of such microbes. Naturally, the credit for this discovery goes to him.

Important stages in agricultural development

Green revolution

Problems of population explosion were started to appear at the beginning of 20th century. Almost all the countries, especially underdeveloped and developing countries had been badly affected by the effects of poor quality and quantity of food. Various methods applied for harvesting maximum yield from minimum land are collectively called as green revolution.



Dr. M.S. Swaminathan



Dr. Norman Borlaug

Improved dwarf varieties of wheat and rice, proper use of fertilizers and pesticides and water management has led to the increased production of food grains and thereby large population had been saved from hunger. Dr. Norman Borlaug (USA) and Dr. M. S. Swaminathan (India) have valuable contribution in green revolution.

Various research institutes and laboratories are engaged in development of new varieties of various crops through research. Ex. Indian Agricultural Research Institute (IARI), New Delhi, National Citrus Research Institute, Nagpur and allied branches, Indian Institute of Sciences, National Pomegranate Research Institute, Solapur.



Collect Information

Which new species of the rice have been developed in India?



Internet is My Friend

Collect the information and make the chart about the work of various state and national-level institutes related with biotechnology.

White revolution

Various parts of India were rich in milk and milk products. However, those products were not sufficient to meet the needs of far-flung regions. Dr. Verghese Kurien proved through the cooperative movement and use of biotechnology that Dairy cannot be allied but it will be a mainstream business. He put the cooperative dairy movement of Anand, Gujarat at all time high status.

While achieving the self-sufficiency in dairy business, various experiments were performed for quality control, newer dairy products and their preservation. Why people from all over the world are again preferring the local wild varieties?

Blue revolution

Production of various useful aquatic organisms with the help of water is called as blue revolution. Farm ponds and the fishes are very common in East Asian countries. However, people are not only thinking of cultivating the fishes and shrimps but other aquatic plants and animals too. Government of India has vowed to increase the production by encouraging the people for pisciculture by launching the program 'Nil-Kranti Mission-2016' (NKM-16). 50% to 100% subsidies are offered in this case.

Marine and fresh water fishery is possible on large scale. Fresh water fishes like rohu, catla and other fishery products like shrimp and lobsters are being cultured on large scale



8.8 Milk processing and milk products



8.9 Pisciculture : Prawns

Fertilizers

Two types of fertilizers are used in agriculture. One of those is organic manure and others are chemical fertilizers. Water holding capacity of the soil improves with soil conservation due to use of manures.

Upper layer of the soil essential in agriculture is formed due to humus formation. Various essential elements like N, P, K can be available to crops due to earthworms and fungi. In soil-less farming i.e. hydroponics, liquid chemical fertilizers are used. However, there are more harmful effects of liberal use of chemical fertilizers. It includes decrease in fertility of soil.

Insecticides

Though the natural immunity of plants can prevent the infections, use of insecticides is not under control. Irrespective of the natural friends of farmers like frogs and insectivorous birds, pesticides are used on large scale for increase in yield. Pesticides are in fact a type of poison. This poison enters the food-web through water and food and its bio-magnification occurs. Various pesticides like DDT, malathion, chloropyrifos, etc. have been proved to be dangerous.

Organic farming

Now a days, organic farming and organic products have become buzzwords. Organic products are being available and demand for them is ever-increasing.

Chemical fertilizers and pesticides have been used on large scale. These poisonous chemicals reached the human body through food and water and their adverse harmful effects on human and environment became apparent.

Various problems like soil fertility and pest infestation have become serious. So as to overcome these problems, farmers are opting for organic farming. It includes complete ban on chemical fertilizers and pesticides and use of local, sturdy varieties and thereby maintaining natural balance. Definitely, this is a welcome decision.

Apiculture

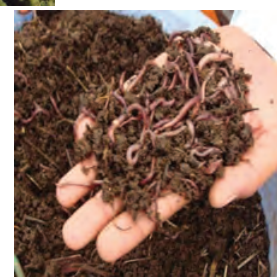
You must have seen the bee hive. Worst method of harvesting / collecting the honey from these hives is driving away the bees by smoking the hive with burning torch and then cutting the hive into pieces. This method causes the destruction of hive and large scale death of bees. However, it is easy to collect the honey without destroying the hive and bees, if artificial bee boxes are used.



8.10 Fertilizers and Insecticides



Oil cake



Vermiculture

8.11 Oil cake and Vermiculture



8.12 Apiculture

Cultivation of Medicinal Plants

India has been gifted with a great biodiversity. Indian citizens have established the humble and strong relation with the nature. We have a great tradition of ayurveda that cures the diseases with the help of natural sources.



8.13 *Adhatoda vasica* (Adulsa) and *Azadirachta Indica* (Neem)

During earlier days, medicinal plants were collected from the forest. However, due to depletion in forest area, medicinal plants are becoming rare. Hence, medicinal plants are being cultivated.



Make a list and discuss

Which fruit processing industries you observe in your surrounding ? What is their effect ?

Fruit Processing

We are daily using various products prepared from fruits. All are consuming the products like chocolates, juices, jams and jellies. All these products can be produced by processing on fruits. Fruits are perishable agro-produce. It needs the processing in such a way that it can be used throughout the year. Fruit processing includes various methods ranging from storage in cold storage to drying, salting, air tight packing, preparing murabba, evaporating, etc.



8.14 Mango processing Unit

Exercise

1. Fill in the blanks and complete the statements.

- Methods like artificial insemination and embryo transplant are mainly used for --
- is the revolutionary event in biotechnology after cloning.
- The disease related with the synthesis of insulin is ---
- Government of India has encouraged the --- for improving the productivity by launching NKM-16.

2. Match the pairs.

- | | |
|-----------------|--------------------|
| a. Interferon | 1. Diabetes |
| b. Factor | 2. Dwarfness |
| c. Somatostatin | 3. Viral infection |
| d. Interleukin | 4. Cancer |
| | 5. Haemophilia |

3. Rewrite the following wrong statements after corrections.

- Changes in genes of the cells are brought about in non-genetic technique.
- Gene from *Bacillus thuringiensis* is introduced into soyabean.

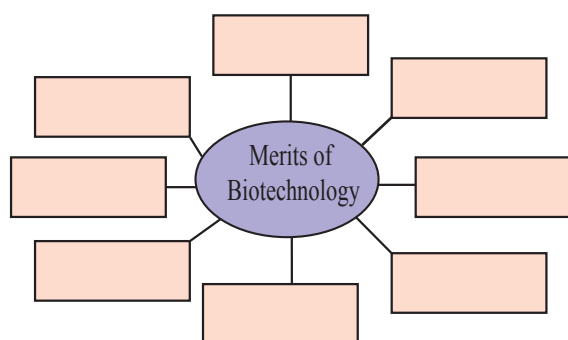
4. Write short notes.

- Biotechnology: Professional uses.
- Importance of medicinal plants.

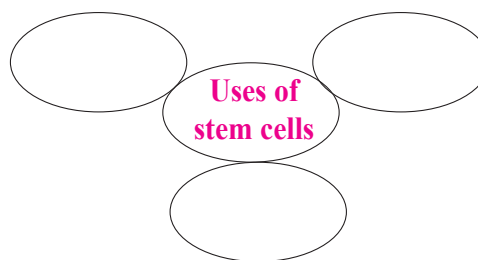
5. Answer the following questions in your own words.

- Which products produced through biotechnology do you use in your daily life?
- Which precautions will you take during spraying of pesticides?
- Why some of the organs in human body are most valuable?
- Explain the importance of fruit processing in human life?
- Explain the meaning of vaccination.

6. Complete the following chart.



7. Write the correct answer in blank circles.



8. Identify and complete the following correlations.

- Insulin : Diabetes :: Interleukin : -- -- --
- Interferon : -- :: Erythropoietin : Anaemia.
- : Dwarfness :: Factor VIII : Haemophilia.
- White revolution : Dairy :: Blue revolution : -- -- -- --

9. Write a comparative note on usefulness and harmfulness of biotechnology.

Projects:

- Visit the organic manuring projects nearby your place and collect more information.
- What will you do to increase public awareness about organ donation in your area?
- Collect information about 'green corridor'. Make a news-collection about it.

